SATA 1D: Random Geometries

Random Heights and Piecewise Ramps, Targets 200 m and 2.6 km Apart

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September 2026

Setup. SATA 1D, $N_{rx} = 4$, PRF = 2000 Hz, $\text{PRF}_{\text{ch}} = 500$ Hz, DPCA along-track baselines ($\Delta b_{at} = 7.69$ m), $b_{xt} = 63.7, 27.0, 4.1, 1.7$ m ($U(0, 100)$, seed 0), $r_0 = 766.2$ km, $\theta = 20°$, $\lambda = 0.25$ m. All targets are iso-range (same slant range r_0, displaced in ground range with their height). Each scene is simulated twice with the *same heights*: targets 200 m apart in azimuth, and targets 2.6 km apart. The 2.6 km spacing is larger than the width of one block of the ΔC_0 map ($2h_w = 2.52$ km), so every target has its own correction, and keeps every pair of targets at least 1 km away from a multiple of 6.23 km, where two targets would share Doppler bins (the Doppler-alias collision); this limits the separated scenes to $N \leq 5$. Each case is reconstructed without SATA, with SATA whole band and with SATA per sub-band.
Scenes. *Random heights*: $N \in \{5, 9, 13\}$ targets, heights $\sim U(0, h_{max})$ with the first target at 0, $h_{max} \in \{100, 500, 1000\}$ m, seeds 0 and 1 (18 cases 200 m apart, 6 cases with $N = 5$ 2.6 km apart). *Piecewise ramp*: $N \in \{5, 9, 13\}$ targets, each step between neighbours with a slope drawn from $\{0.1, 1, 1.5, 2, 3, 4, 5, 10, 15, 20, 30, 40, 50\}°$ with random sign over a 200-m step, seeds 0–4 (15 cases 200 m apart, 5 cases with $N = 5$ 2.6 km apart). The separated scenes keep the heights of the 200-m scenes.

1 The ambiguity metric

The earlier heatmaps used the *worst ambiguity*: the largest value of the focused response outside ±40 samples (±154 m) of every target. That maximum is not always an ambiguity. With SATA it is often the pedestal around the main lobe (the skirt at $|t| \lesssim 0.3$ s, about −36 dB), which does not depend on the correction; without SATA it is the ambiguity at ±0.81 s. In case RA-007 ($N = 2$, $\alpha = 5°$, 200 m apart) both happen to be at −36.5 dB, so the improvement read 0 dB although the ambiguities differ by 12 dB (Fig. 1).

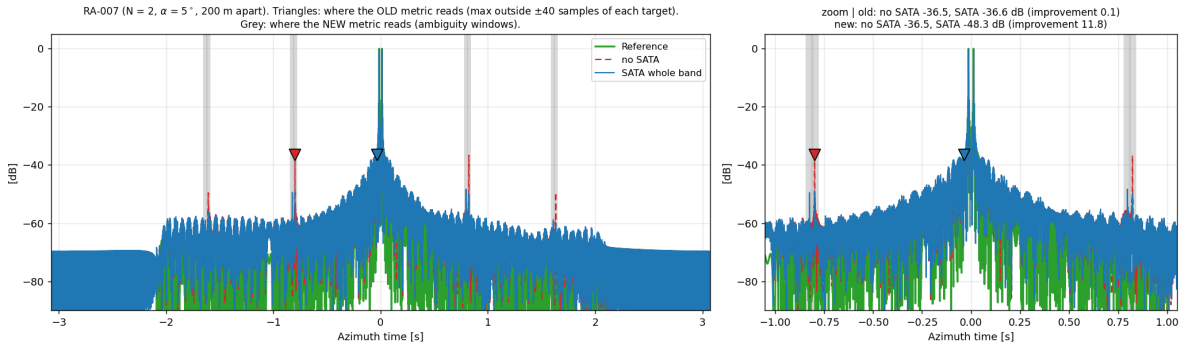


Figure 1: Case RA-007. Triangles: where the old metric reads, for no SATA (the ambiguity at +0.81 s) and for SATA (the main-lobe pedestal). Grey bands: the windows of the metric used here.

Ambiguity peak. A Doppler shift of $m\,\mathrm{PRF_{ch}}$ moves the focused copy of a target by $m\,\Delta t_{amb}$, with

$$\Delta t_{amb} = \frac{\mathrm{PRF_{ch}}}{K_a} = \frac{\mathrm{PRF_{ch}}\,\lambda\, r_0}{2\,v^2} = 0.810 \text{ s} \quad (6.23 \text{ km}). \tag{1}$$

The ambiguity peak is the largest |IRF| within ±20 ms of $x_j/v + m\,\Delta t_{amb}$, $m = \pm 1, \pm 2$, over all targets j, leaving out ±20 ms around every target's own main lobe, in dB relative to the reference peak. The grey bands in every IRF page show where it is read. Improvement = ambiguity peak (no SATA) − ambiguity peak (SATA). For the reference, which has no ambiguities, it reads the level of the IRF floor in the windows (−52 to −63 dB); with targets 2.6 km apart and $N \geq 3$ the windows of one target also see the skirt of the main lobe of a neighbour, which sets a floor of −46 dB ($N = 3$) and −43 dB ($N = 5$) common to all methods.

2 Results

family	spacing	cases	no SATA median	SATA whole band median (range)	improvement median (range)	|sub − whole| max
random heights	200 m	18	−7.9 dB	−17.8 (−33.8 . . . − 11.9) dB	9.2 (4.8–12.8) dB	0.04 dB
	2.6 km	6	−9.3 dB	−43.1 (−43.3 . . . − 43.0) dB	33.9 (18.6–36.7) dB	0.22 dB
piecewise ramp	200 m	15	−14.1 dB	−28.3 (−40.6 . . . − 24.5) dB	14.5 (7.0–20.7) dB	0.07 dB
	2.6 km	5	−17.0 dB	−43.2 (−43.2 . . . − 43.0) dB	26.2 (11.4–30.5) dB	0.12 dB

Table 1: Ambiguity peak, medians over the cases. For $N = 5$ at 2.6 km, −43 dB is the floor of the metric (the skirt of a neighbour's main lobe in the ambiguity windows, the same for all methods).

- **Targets 200 m apart.** SATA improves every case, by 4.8 to 20.7 dB, but the ambiguity after SATA follows the height differences between neighbours: −31 to −34 dB for $h_{max} = 100$ m, −17 to −18 dB for 500 m and −12 to −13 dB for 1000 m (random heights). Neighbours 200 m apart fall into one sub-aperture cell ($\delta x = 309$ m) and receive one common correction; random terrain with hundreds of metres between neighbours is exactly the case this cannot resolve.
- **Targets 2.6 km apart, same heights.** The ambiguity after SATA is at the −43 dB floor of the metric in every case, whatever the heights (up to 1000 m) or the slopes (up to ±50° between neighbours in the 200-m version): each target has its own block in the ΔC_0 map and its own correction. The improvement rises to 18.6–36.7 dB (random heights) and 11.4–30.5 dB (piecewise).
- **Whole band and per sub-band** agree within 0.22 dB in every case.
- **Spacing** (Fig. 2). With the heights of one random scene fixed, the ambiguity after SATA drops steadily as the targets move apart, except at the collision spacings 1.6 and 2 km.

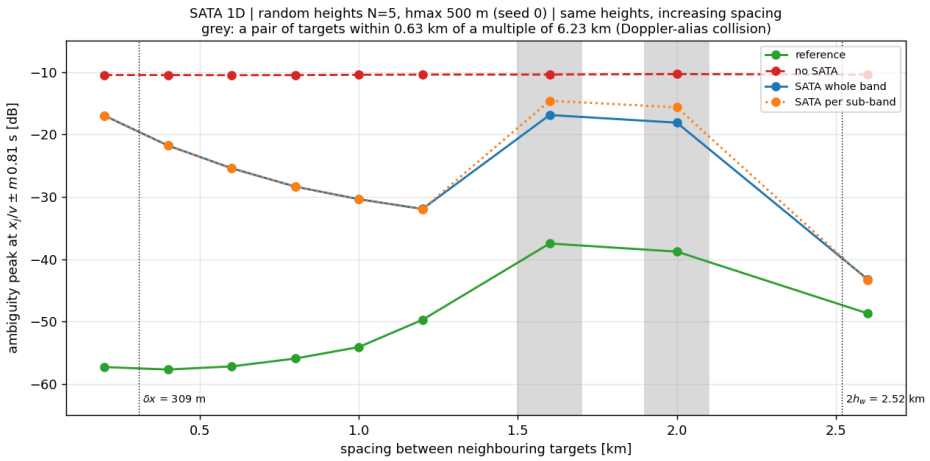


Figure 2: Random heights $N = 5$, $h_{max} = 500$ m, seed 0: ambiguity peak against the spacing between neighbouring targets. Grey: collision spacings; there the ambiguity of one target falls 20–30 ms from the main lobe of another target, so even the reference reads −38 dB in the windows.

3 Heatmaps

For each family and spacing, two pages: the improvement over no SATA (SATA whole band and SATA per sub-band), and the ambiguity peak itself (no SATA, SATA whole band, SATA per sub-band).

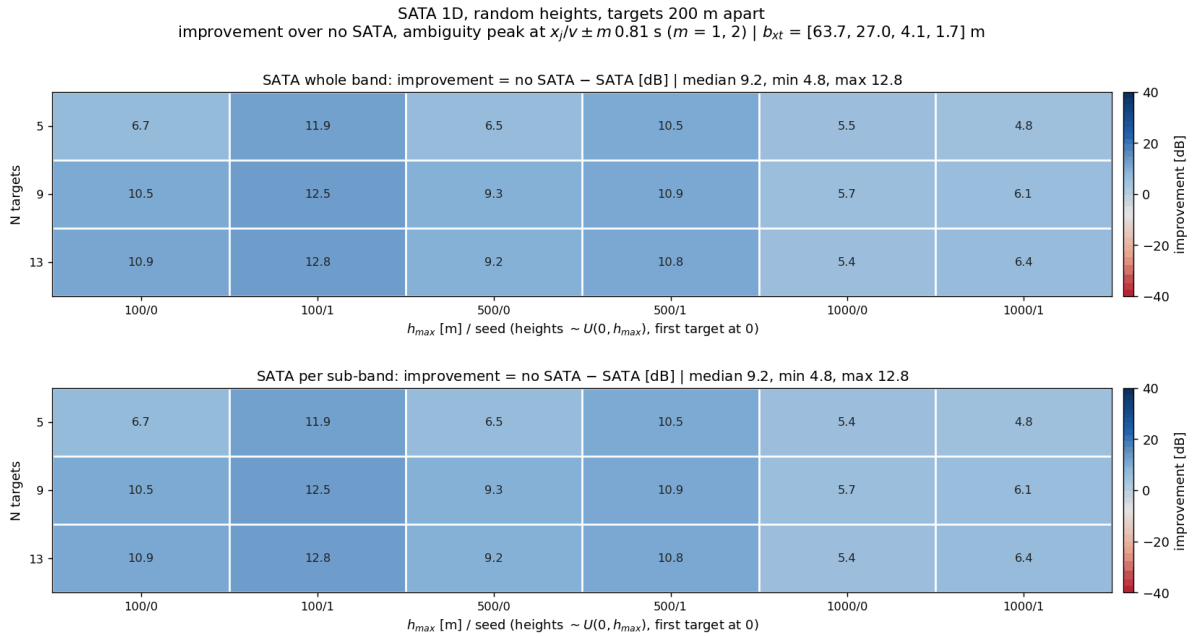


Figure 3: Random heights, targets 200 m apart. Columns: h_{max} / seed. Improvement over no SATA, whole band (top) and per sub-band (bottom).



Figure 4: Random heights, targets 200 m apart. Columns: h_{max} / seed. Ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

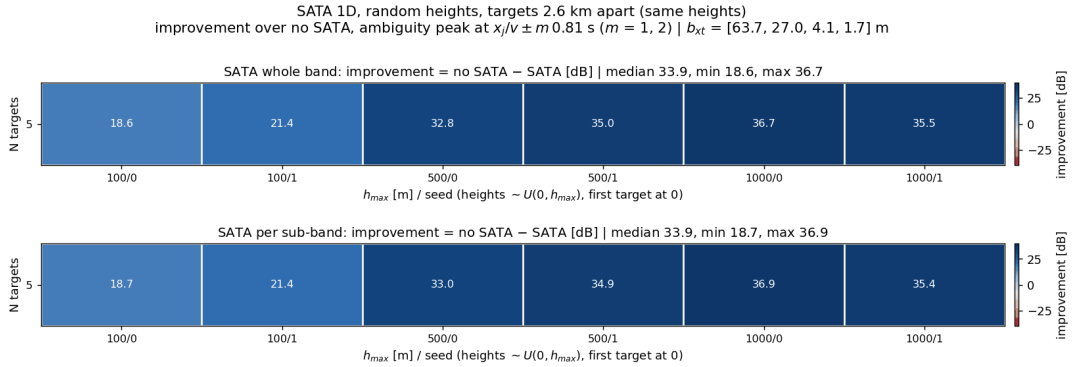


Figure 5: Random heights, targets 2.6 km apart, same heights. Columns: h_{max} / seed. Improvement over no SATA, whole band (top) and per sub-band (bottom).

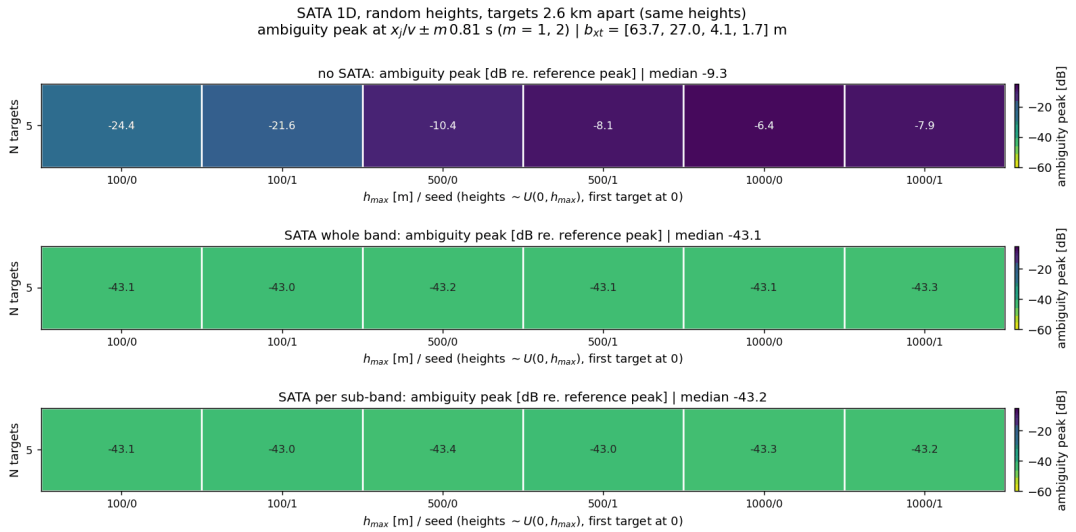


Figure 6: Random heights, targets 2.6 km apart, same heights. Columns: h_{max} / seed. Ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

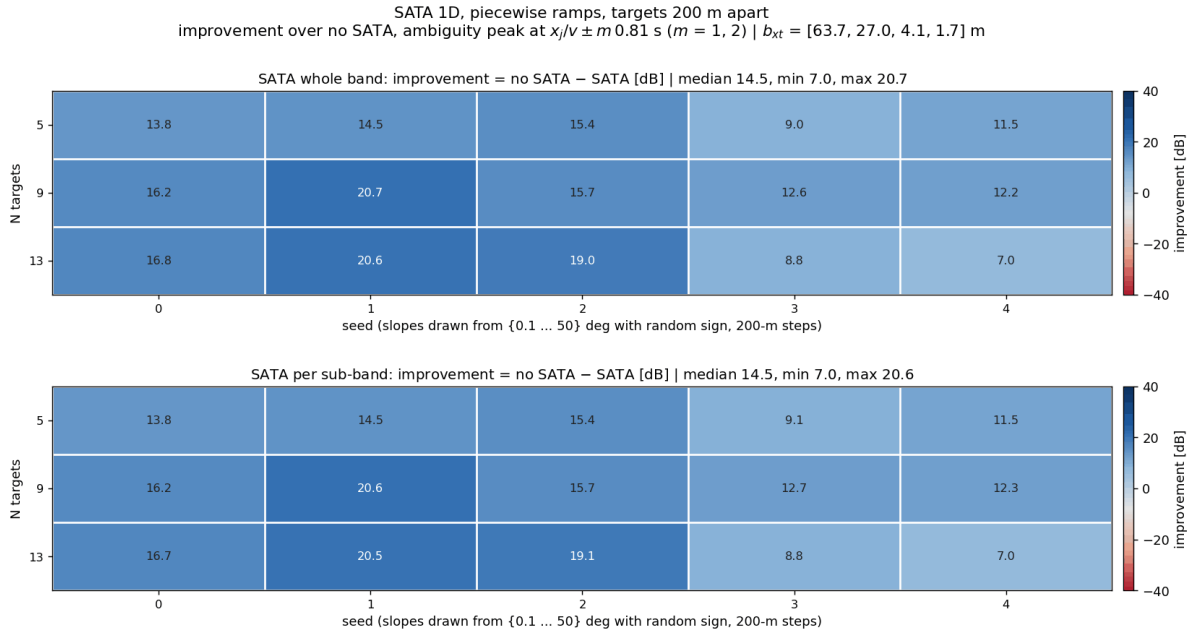


Figure 7: Piecewise ramps, targets 200 m apart. Columns: seed. Improvement over no SATA, whole band (top) and per sub-band (bottom).



Figure 8: Piecewise ramps, targets 200 m apart. Columns: seed. Ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

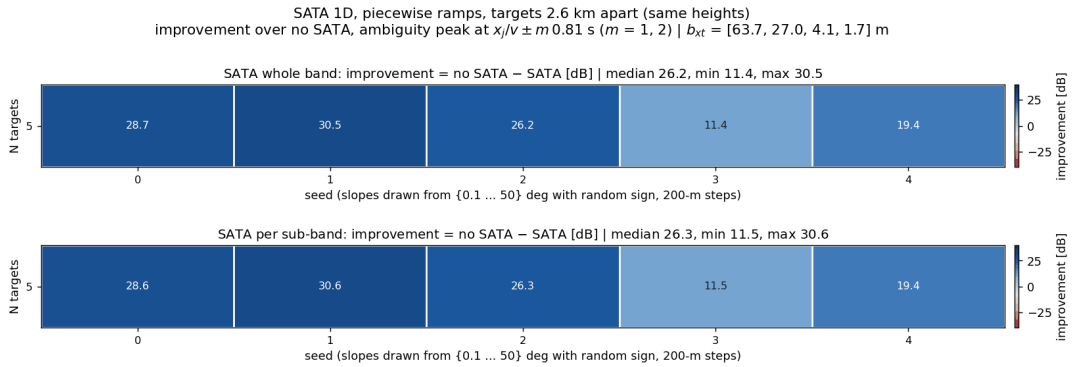


Figure 9: Piecewise ramps, targets 2.6 km apart, same heights. Columns: seed. Improvement over no SATA, whole band (top) and per sub-band (bottom).

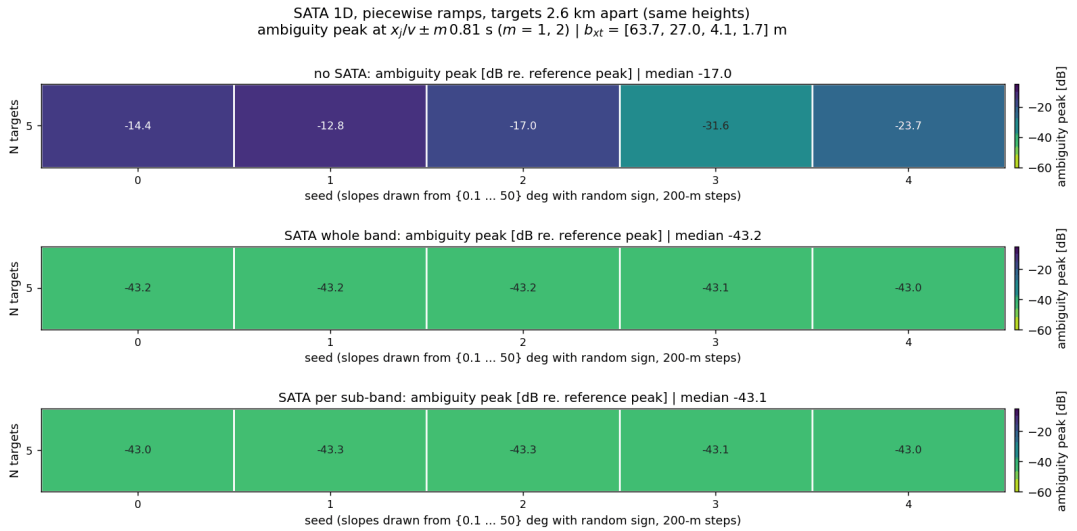


Figure 10: Piecewise ramps, targets 2.6 km apart, same heights. Columns: seed. Ambiguity peak, no SATA (top), SATA whole band (middle), SATA per sub-band (bottom).

4 IRFs, random heights, targets 200 m apart

Top: reference, SATA whole band and SATA per sub-band. Bottom: reference and no SATA. Grey bands: the ambiguity windows. Right: target geometry in 2D and in 3D (azimuth, ground-range offset, height).

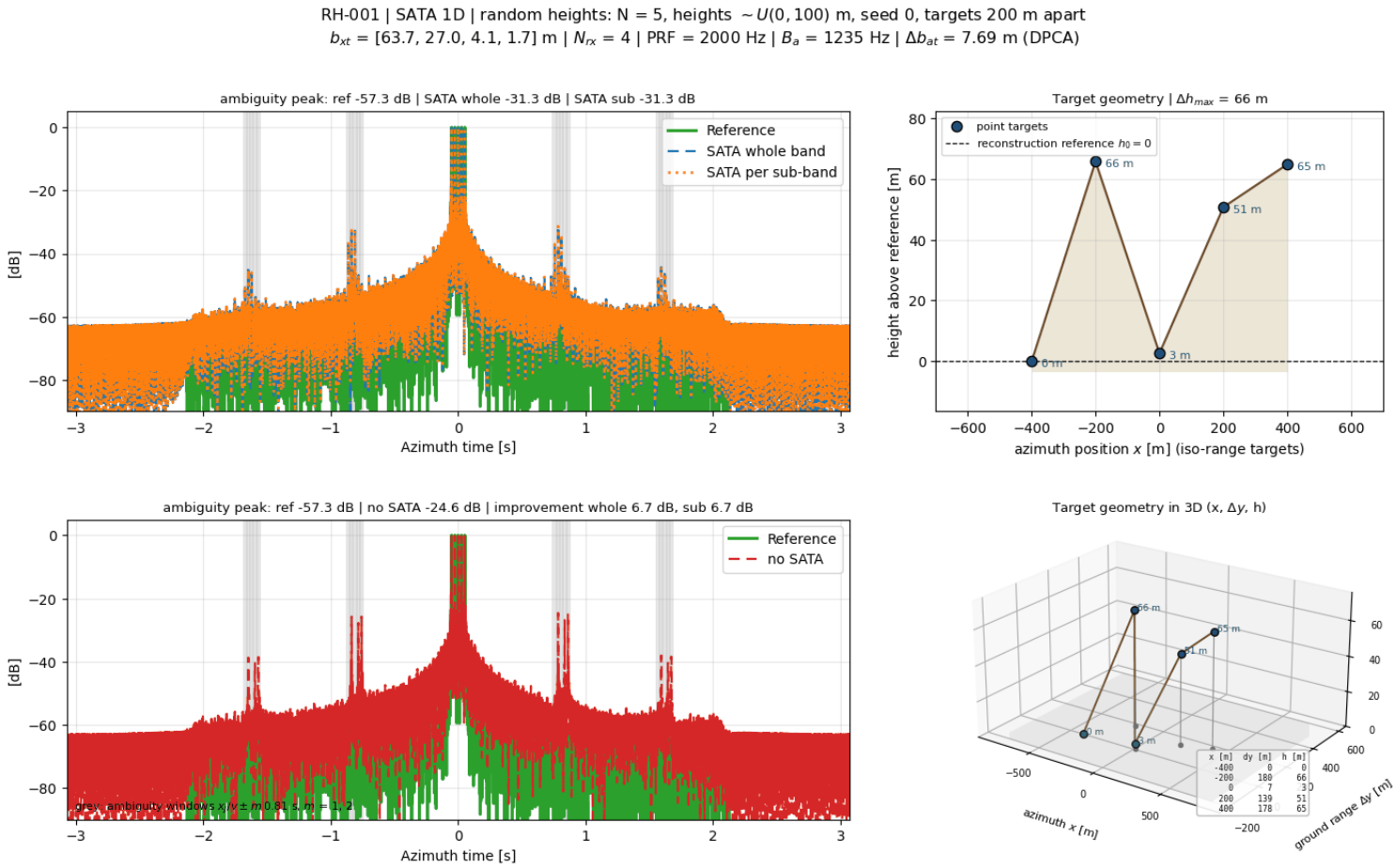


Figure 11: Random heights, 200 m apart: RH-001, $N = 5$, $h \sim U(0, 100)$ m, seed 0. Ambiguity peak: no SATA -24.6, SATA whole band -31.3, per sub-band -31.3 dB (improvement 6.7 dB).

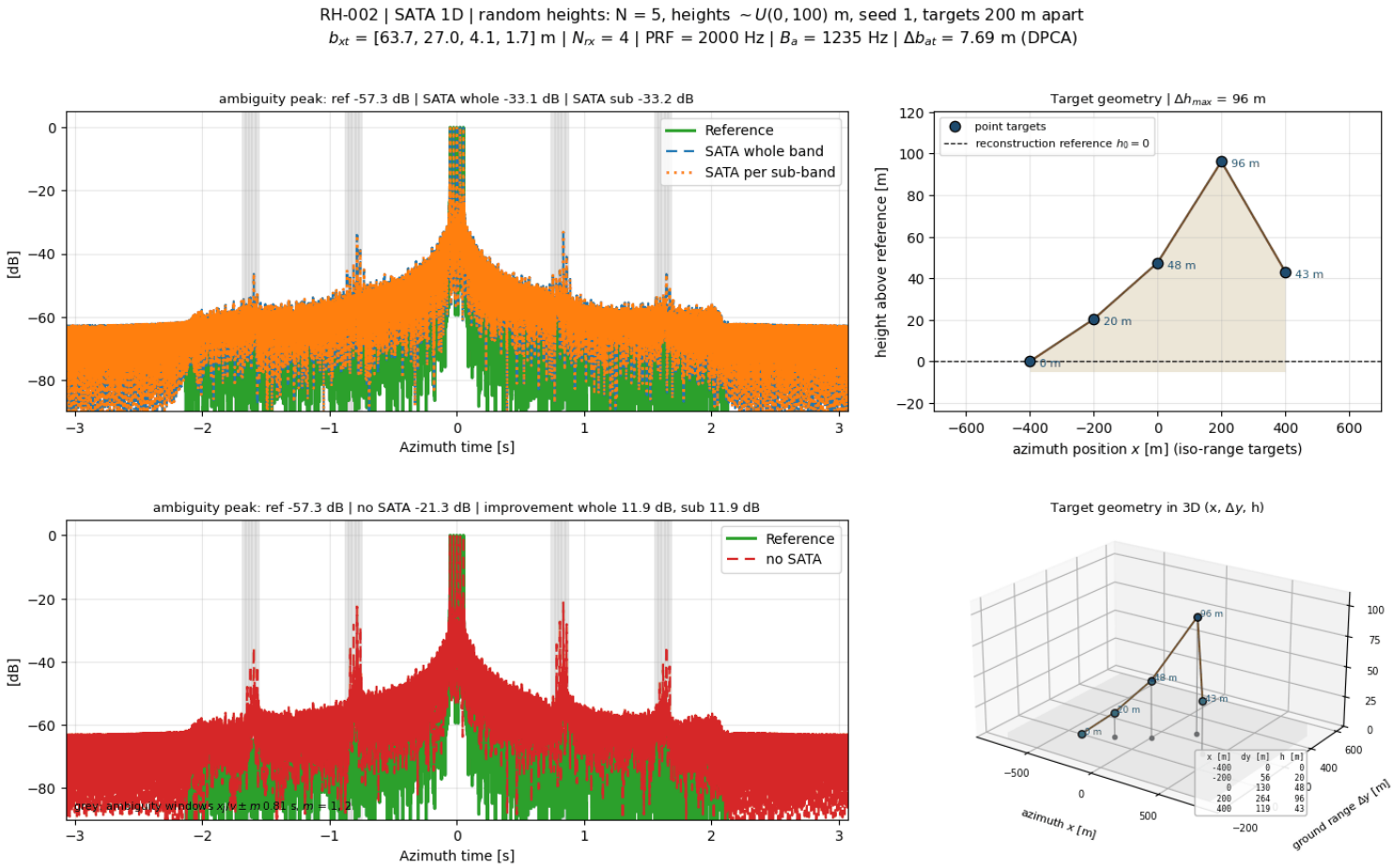


Figure 12: Random heights, 200 m apart: RH-002, $N = 5$, $h \sim U(0, 100)$ m, seed 1. Ambiguity peak: no SATA -21.3, SATA whole band -33.1, per sub-band -33.2 dB (improvement 11.9 dB).

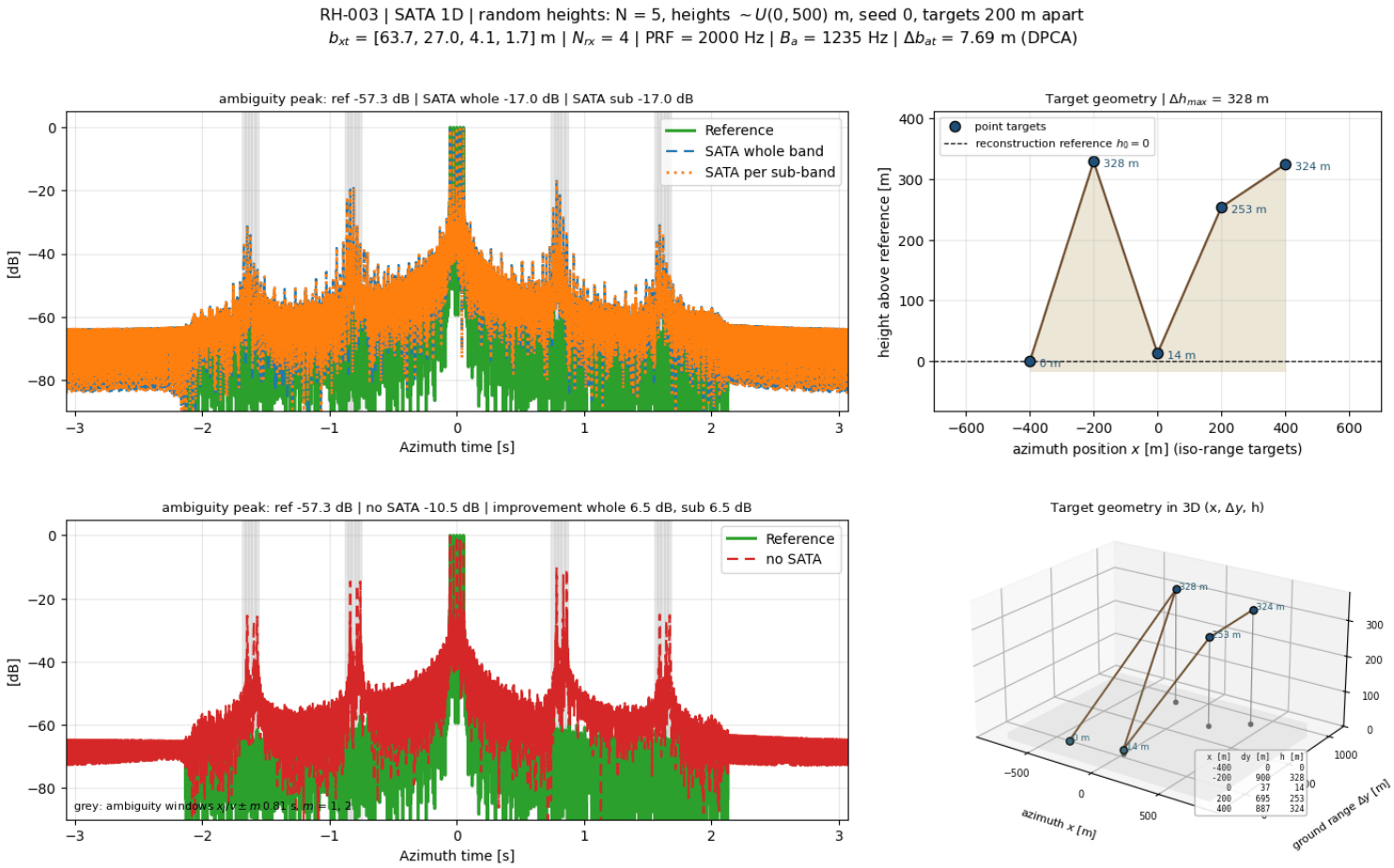


Figure 13: Random heights, 200 m apart: RH-003, $N = 5$, $h \sim U(0, 500)$ m, seed 0. Ambiguity peak: no SATA -10.5, SATA whole band -17.0, per sub-band -17.0 dB (improvement 6.5 dB).

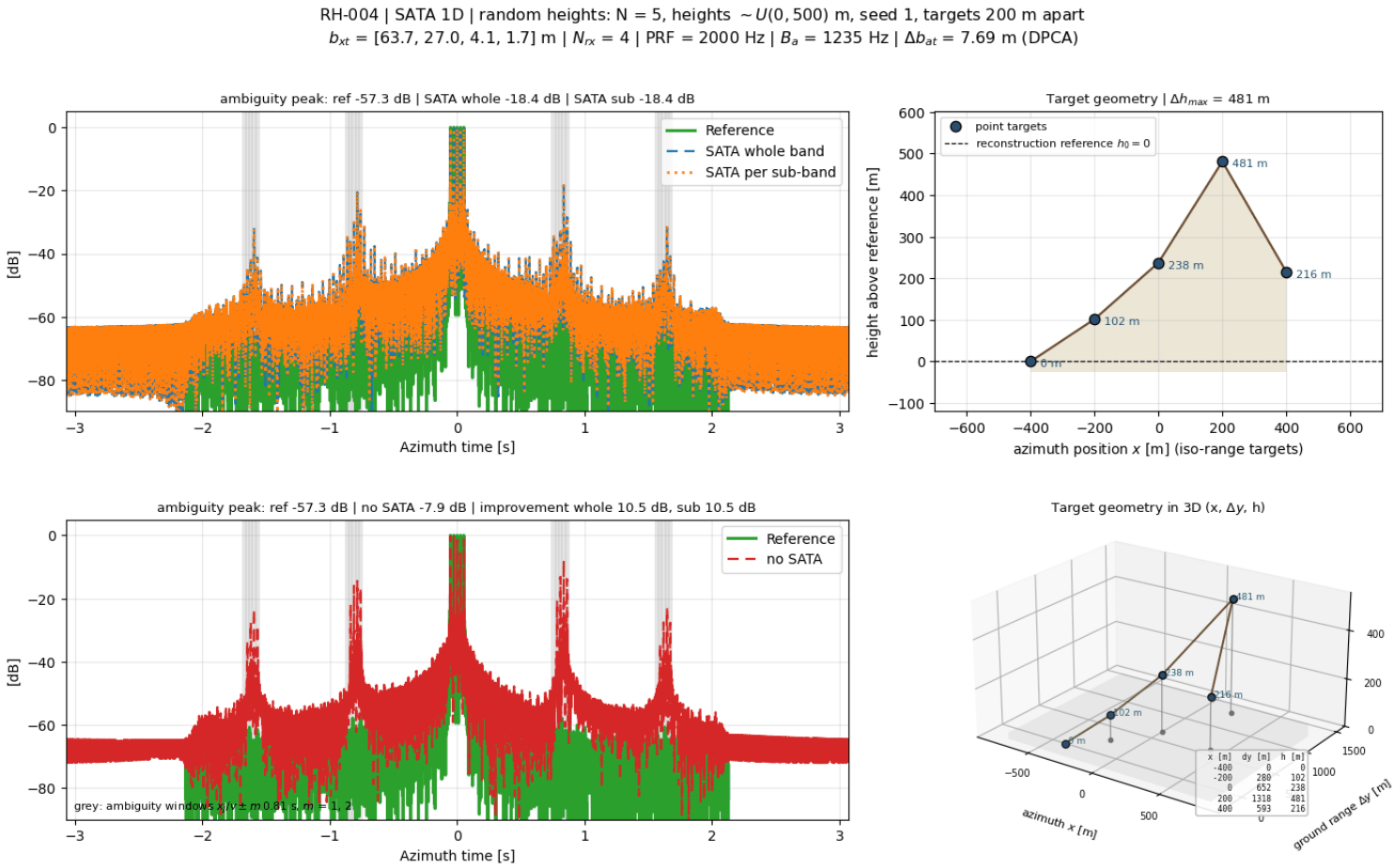


Figure 14: Random heights, 200 m apart: RH-004, $N = 5$, $h \sim U(0, 500)$ m, seed 1. Ambiguity peak: no SATA −7.9, SATA whole band −18.4, per sub-band −18.4 dB (improvement 10.5 dB).

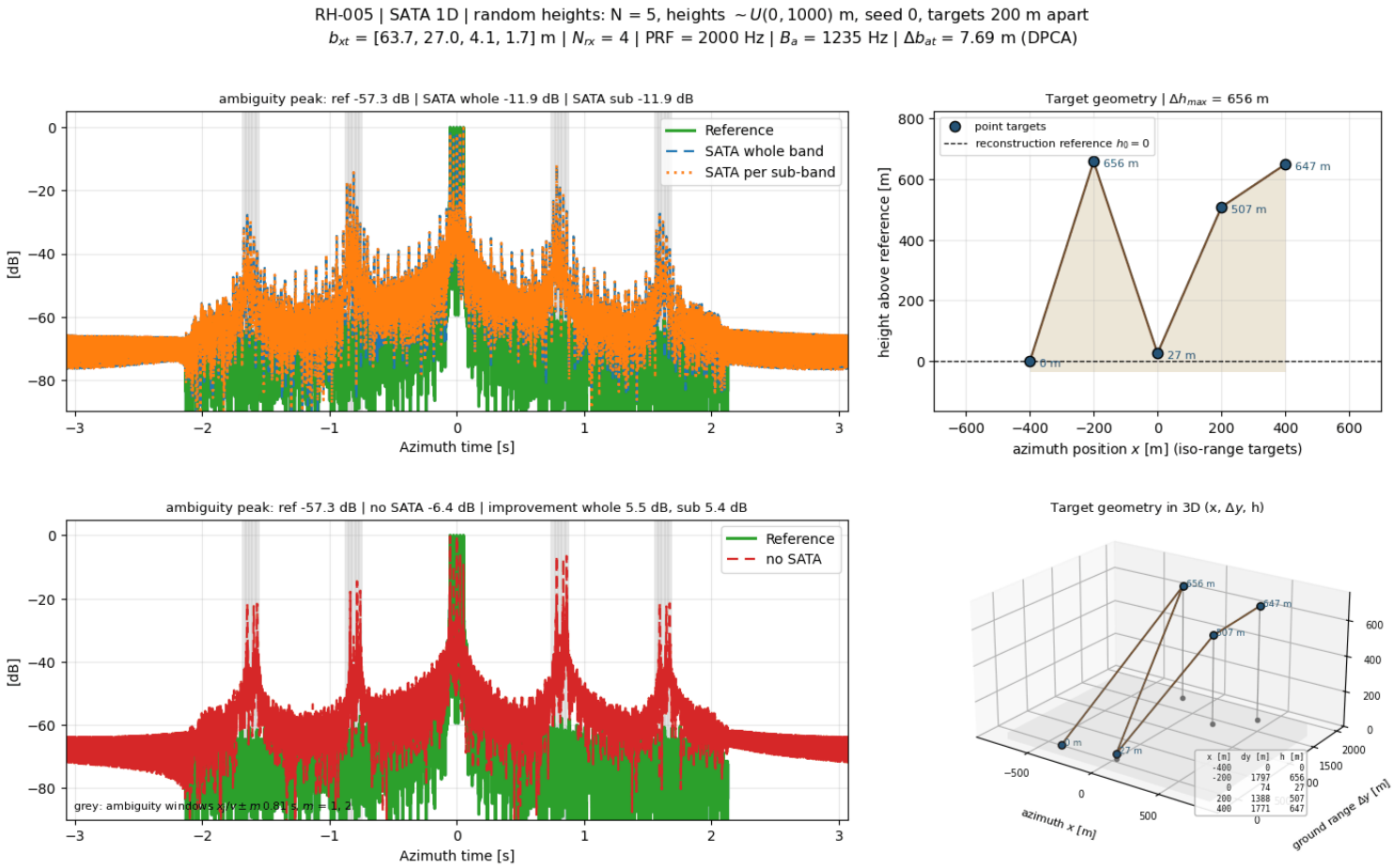


Figure 15: Random heights, 200 m apart: RH-005, $N = 5$, $h \sim U(0, 1000)$ m, seed 0. Ambiguity peak: no SATA -6.4, SATA whole band -11.9, per sub-band -11.9 dB (improvement 5.5 dB).

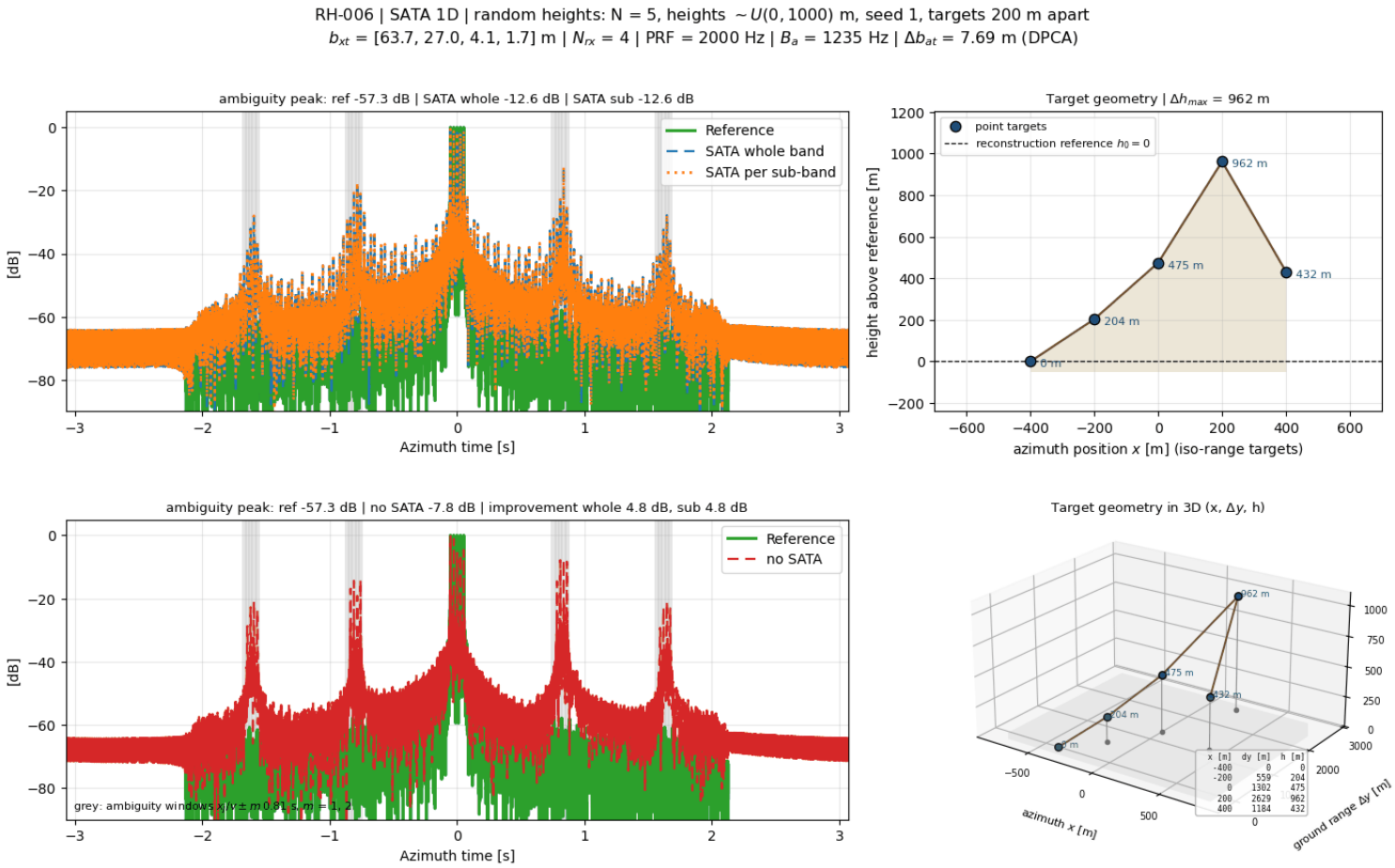


Figure 16: Random heights, 200 m apart: RH-006, $N = 5$, $h \sim U(0, 1000)$ m, seed 1. Ambiguity peak: no SATA -7.8, SATA whole band -12.6, per sub-band -12.6 dB (improvement 4.8 dB).

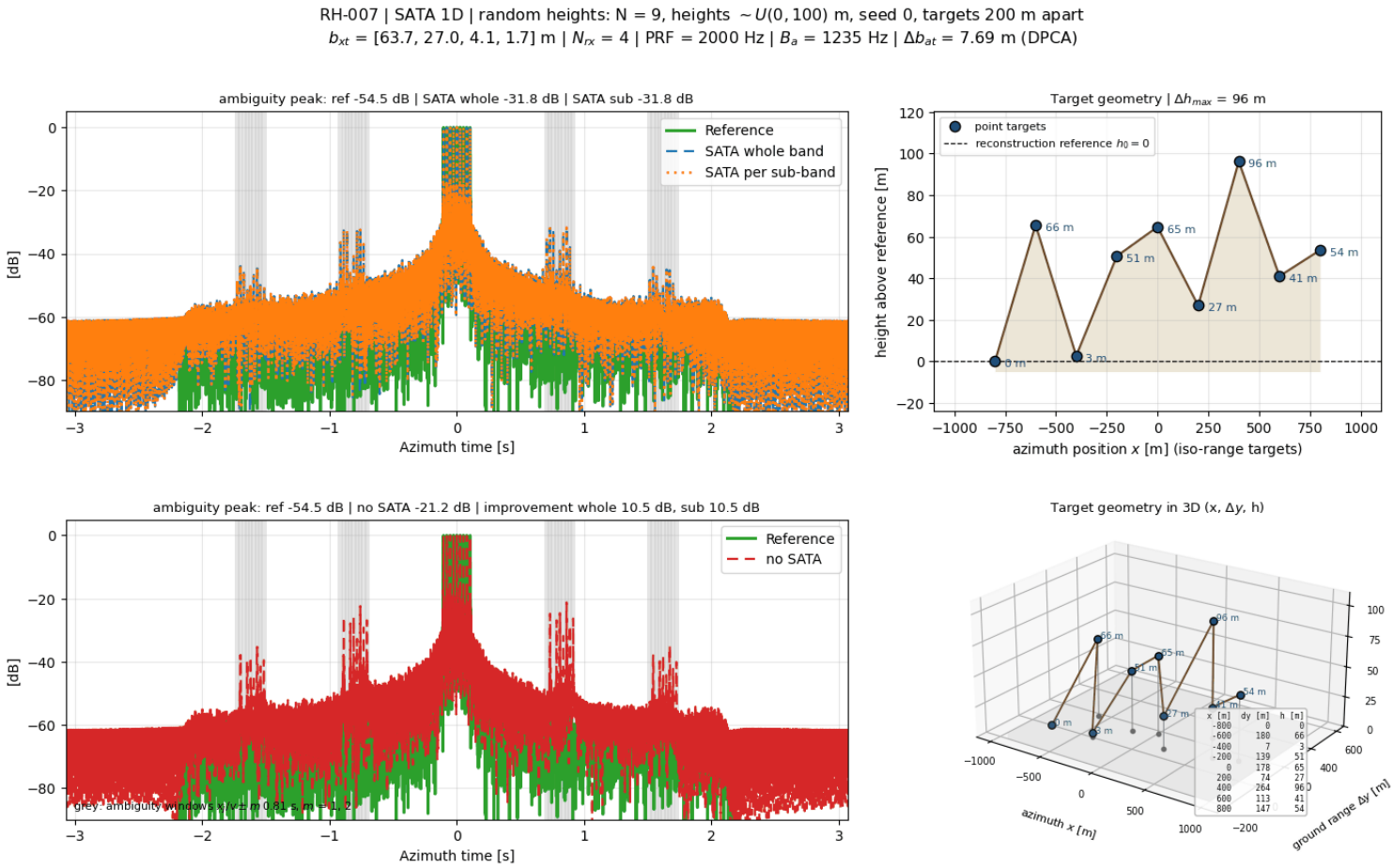


Figure 17: Random heights, 200 m apart: RH-007, $N = 9$, $h \sim U(0, 100)$ m, seed 0. Ambiguity peak: no SATA -21.2, SATA whole band -31.8, per sub-band -31.8 dB (improvement 10.5 dB).

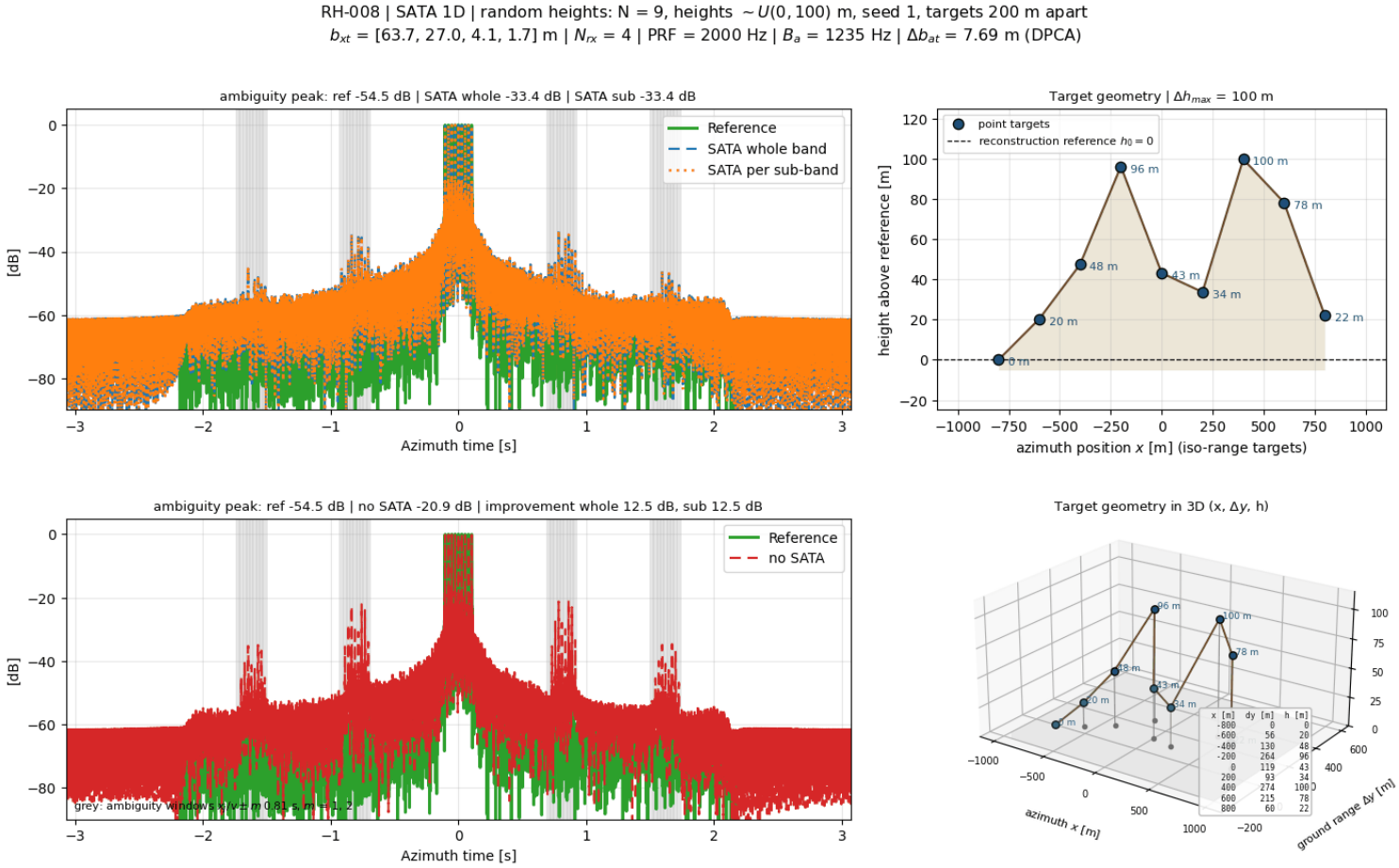


Figure 18: Random heights, 200 m apart: RH-008, $N = 9$, $h \sim U(0, 100)$ m, seed 1. Ambiguity peak: no SATA -20.9, SATA whole band -33.4, per sub-band -33.4 dB (improvement 12.5 dB).

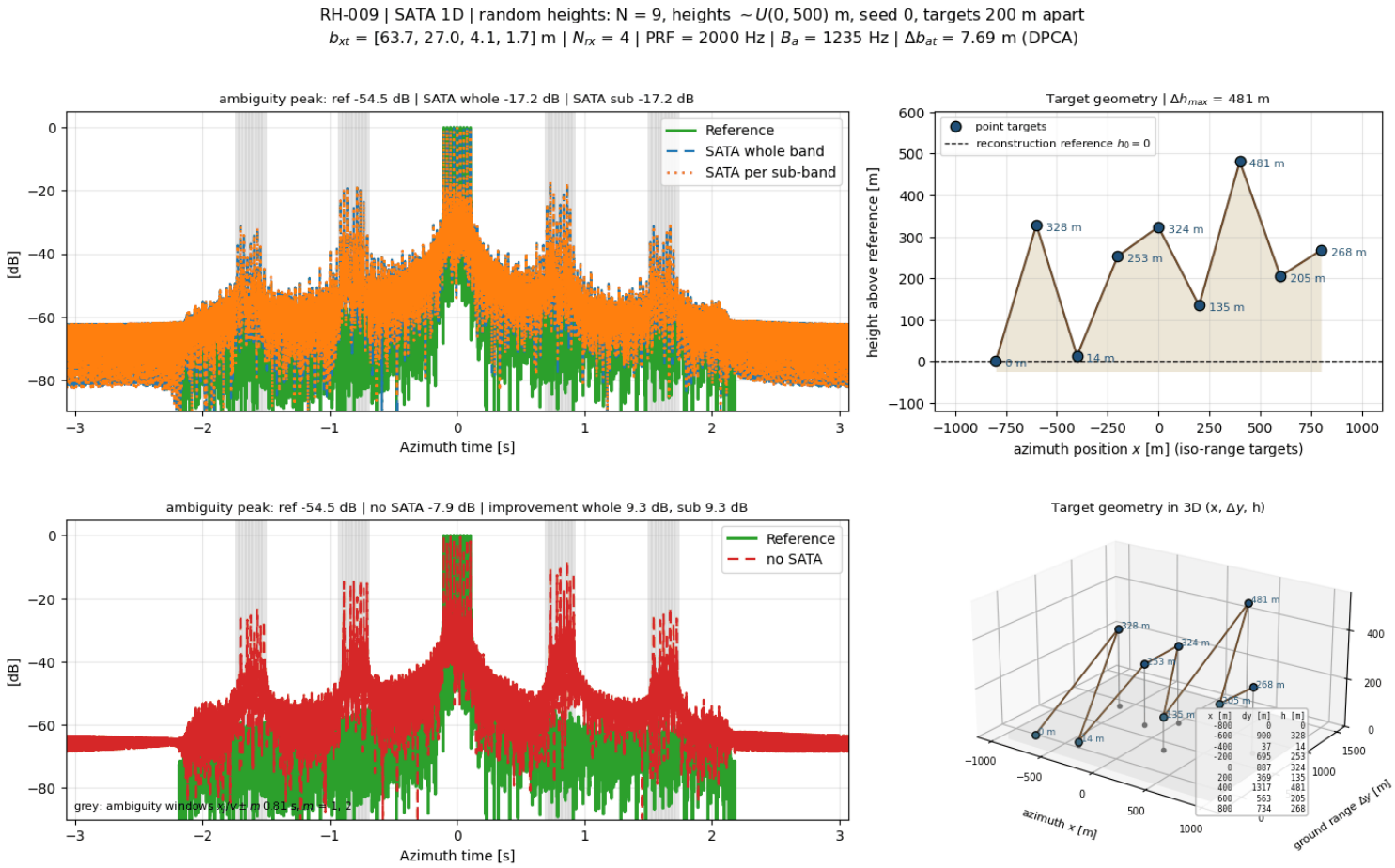


Figure 19: Random heights, 200 m apart: RH-009, $N = 9$, $h \sim U(0, 500)$ m, seed 0. Ambiguity peak: no SATA -7.9, SATA whole band -17.2, per sub-band -17.2 dB (improvement 9.3 dB).

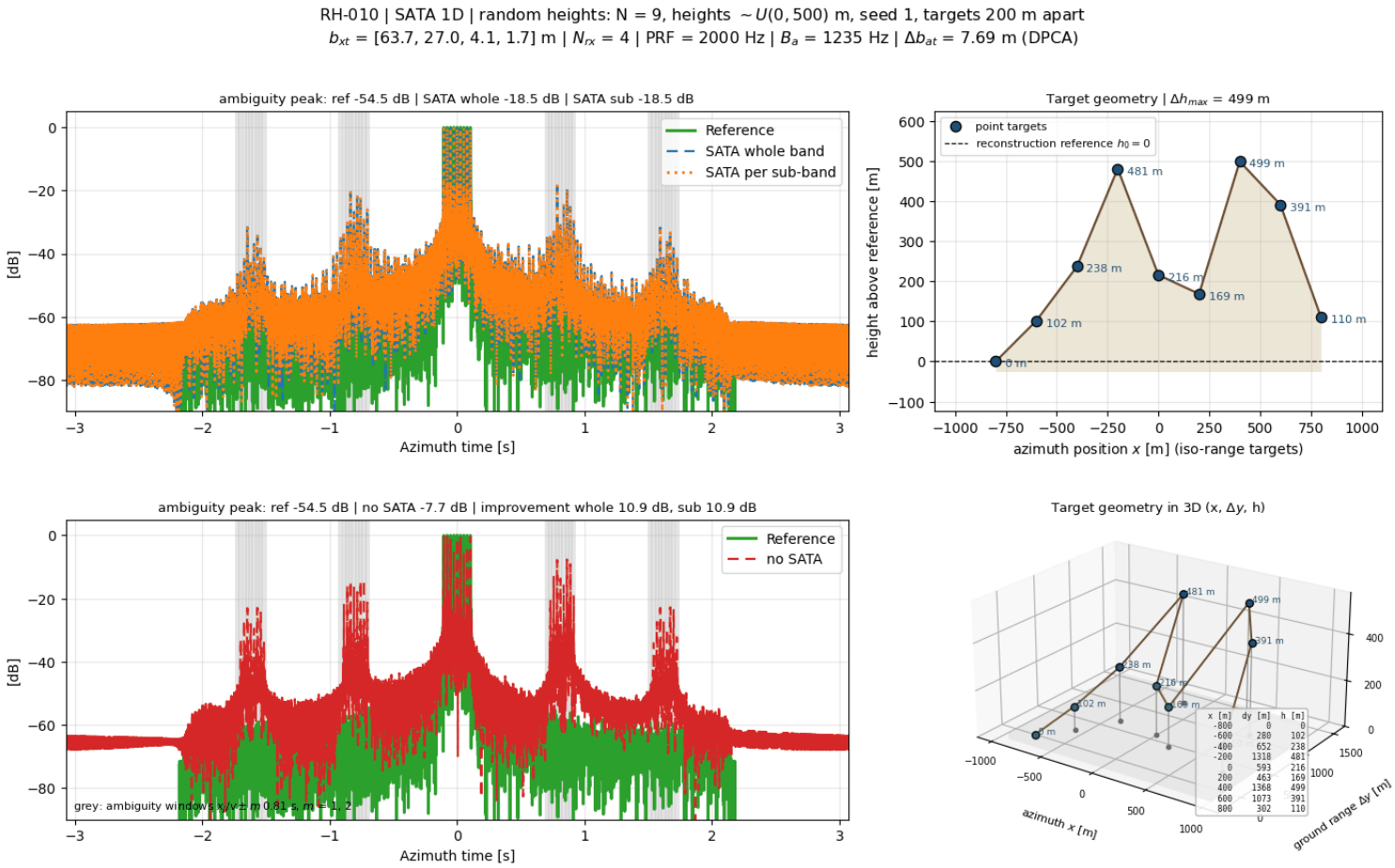


Figure 20: Random heights, 200 m apart: RH-010, $N = 9$, $h \sim U(0, 500)$ m, seed 1. Ambiguity peak: no SATA -7.7, SATA whole band -18.5, per sub-band -18.5 dB (improvement 10.9 dB).

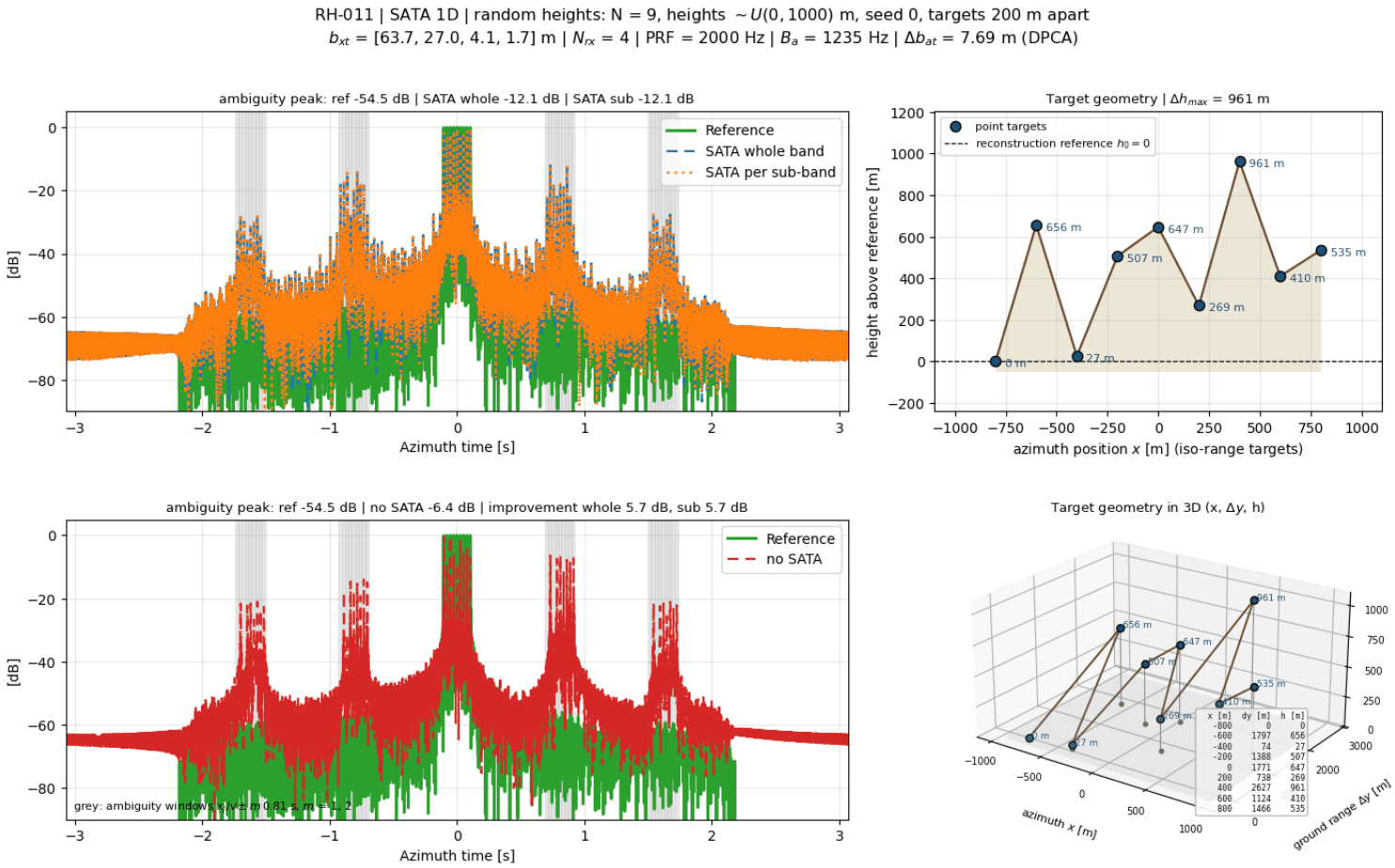


Figure 21: Random heights, 200 m apart: RH-011, $N = 9$, $h \sim U(0, 1000)$ m, seed 0. Ambiguity peak: no SATA -6.4, SATA whole band -12.1, per sub-band -12.1 dB (improvement 5.7 dB).

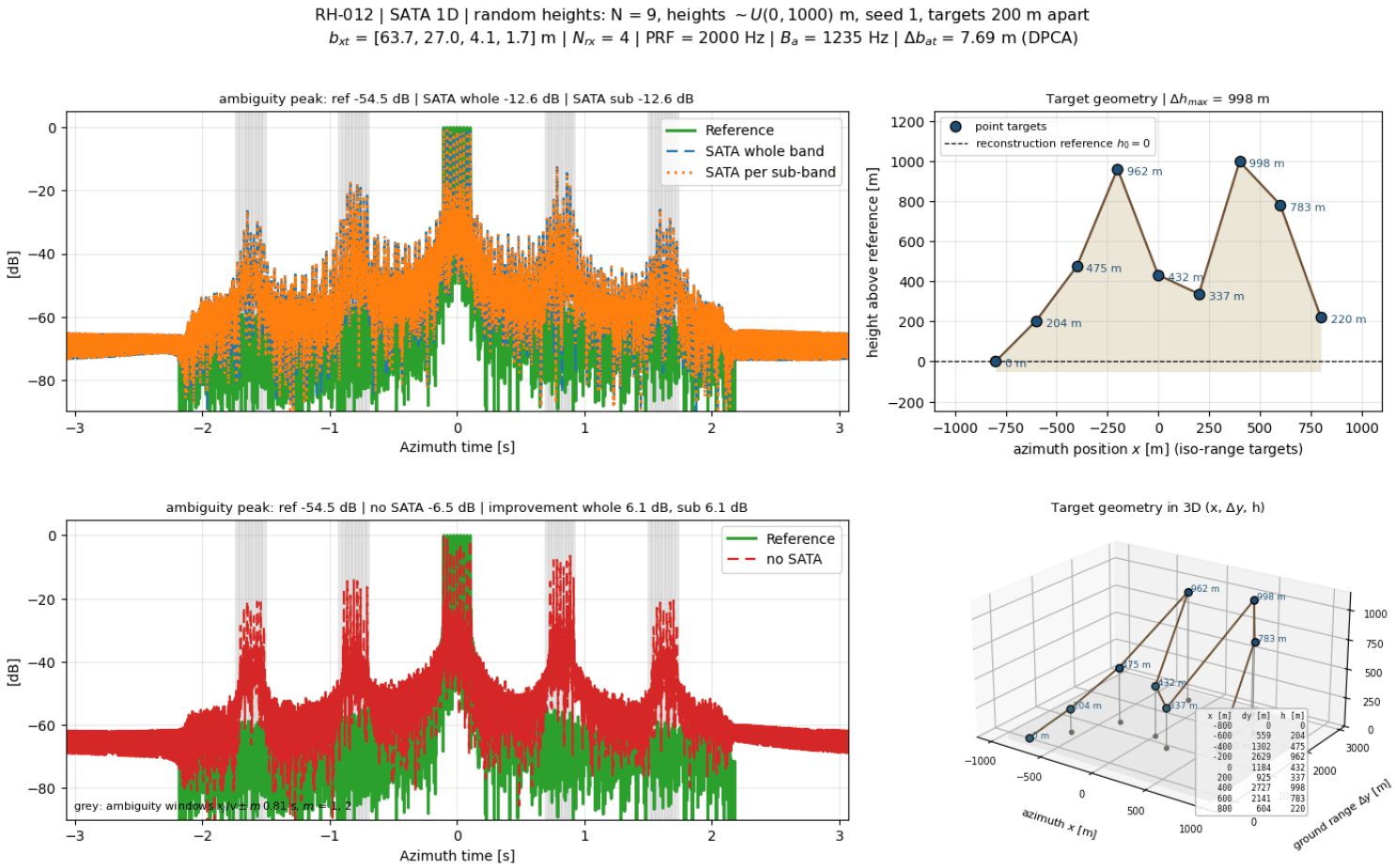


Figure 22: Random heights, 200 m apart: RH-012, $N = 9$, $h \sim U(0, 1000)$ m, seed 1. Ambiguity peak: no SATA -6.5, SATA whole band -12.6, per sub-band -12.6 dB (improvement 6.1 dB).

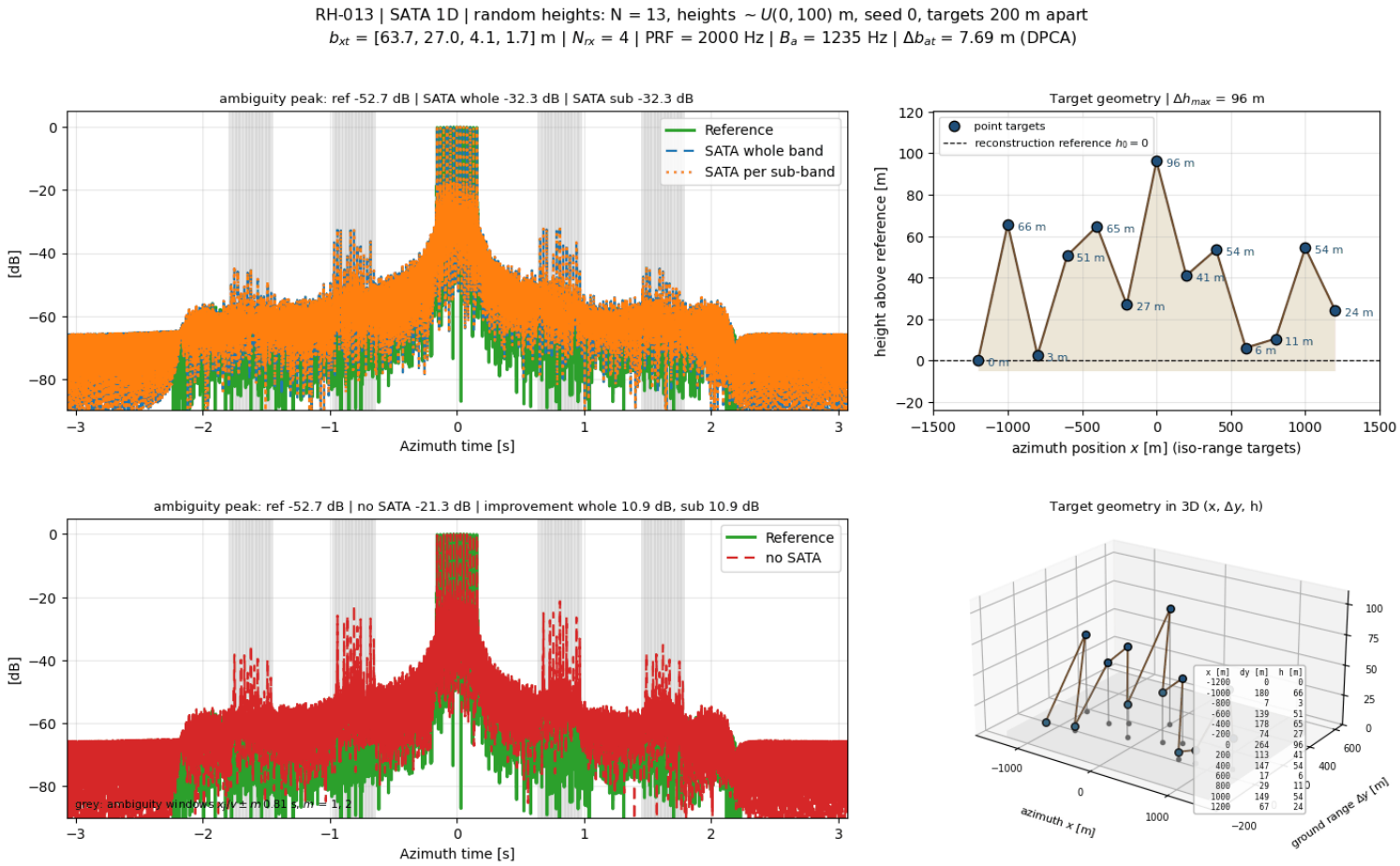


Figure 23: Random heights, 200 m apart: RH-013, $N = 13$, $h \sim U(0, 100)$ m, seed 0. Ambiguity peak: no SATA −21.3, SATA whole band −32.3, per sub-band −32.3 dB (improvement 10.9 dB).

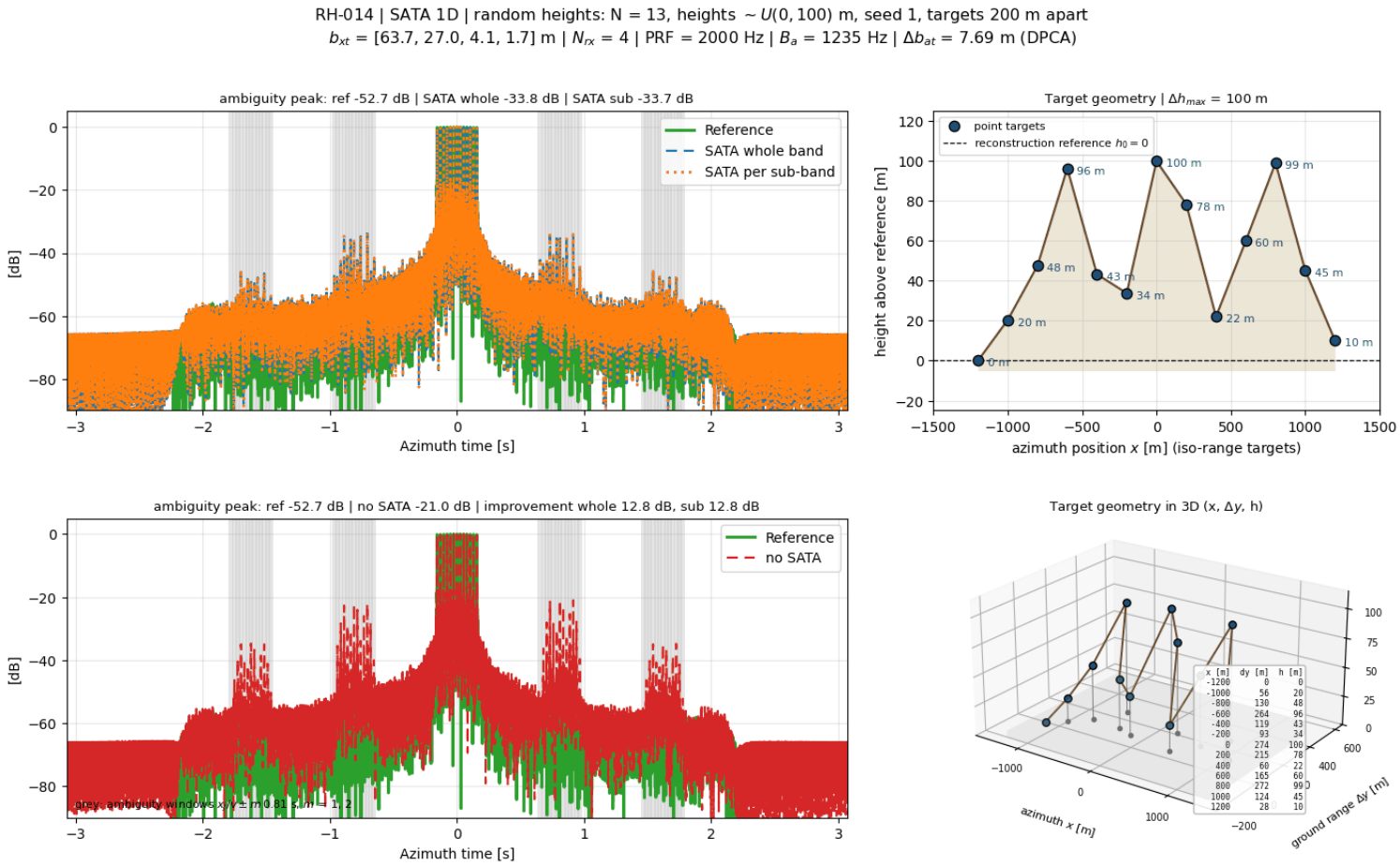


Figure 24: Random heights, 200 m apart: RH-014, $N = 13$, $h \sim U(0, 100)$ m, seed 1. Ambiguity peak: no SATA −21.0, SATA whole band −33.8, per sub-band −33.7 dB (improvement 12.8 dB).

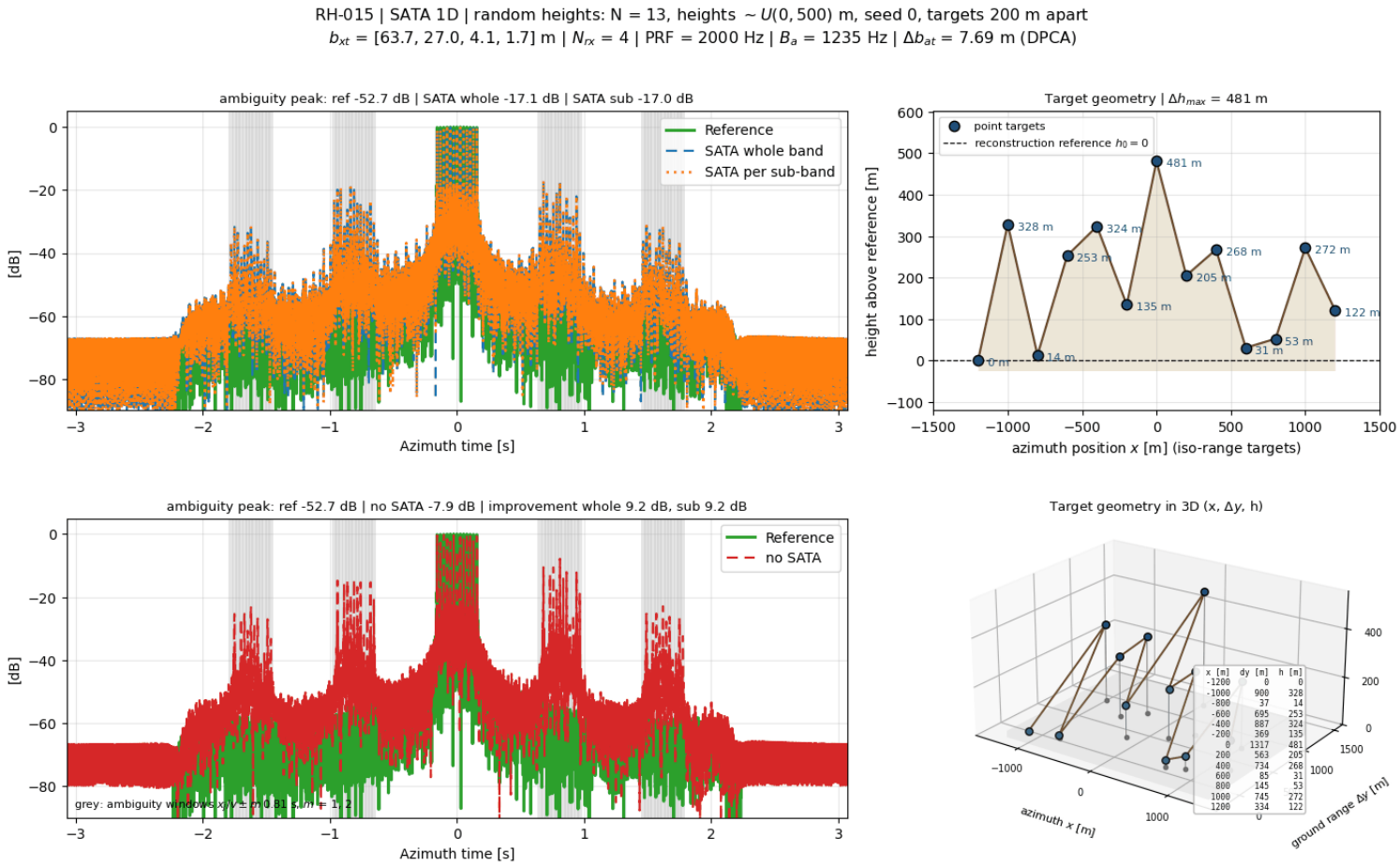


Figure 25: Random heights, 200 m apart: RH-015, $N = 13$, $h \sim U(0, 500)$ m, seed 0. Ambiguity peak: no SATA -7.9, SATA whole band -17.1, per sub-band -17.0 dB (improvement 9.2 dB).

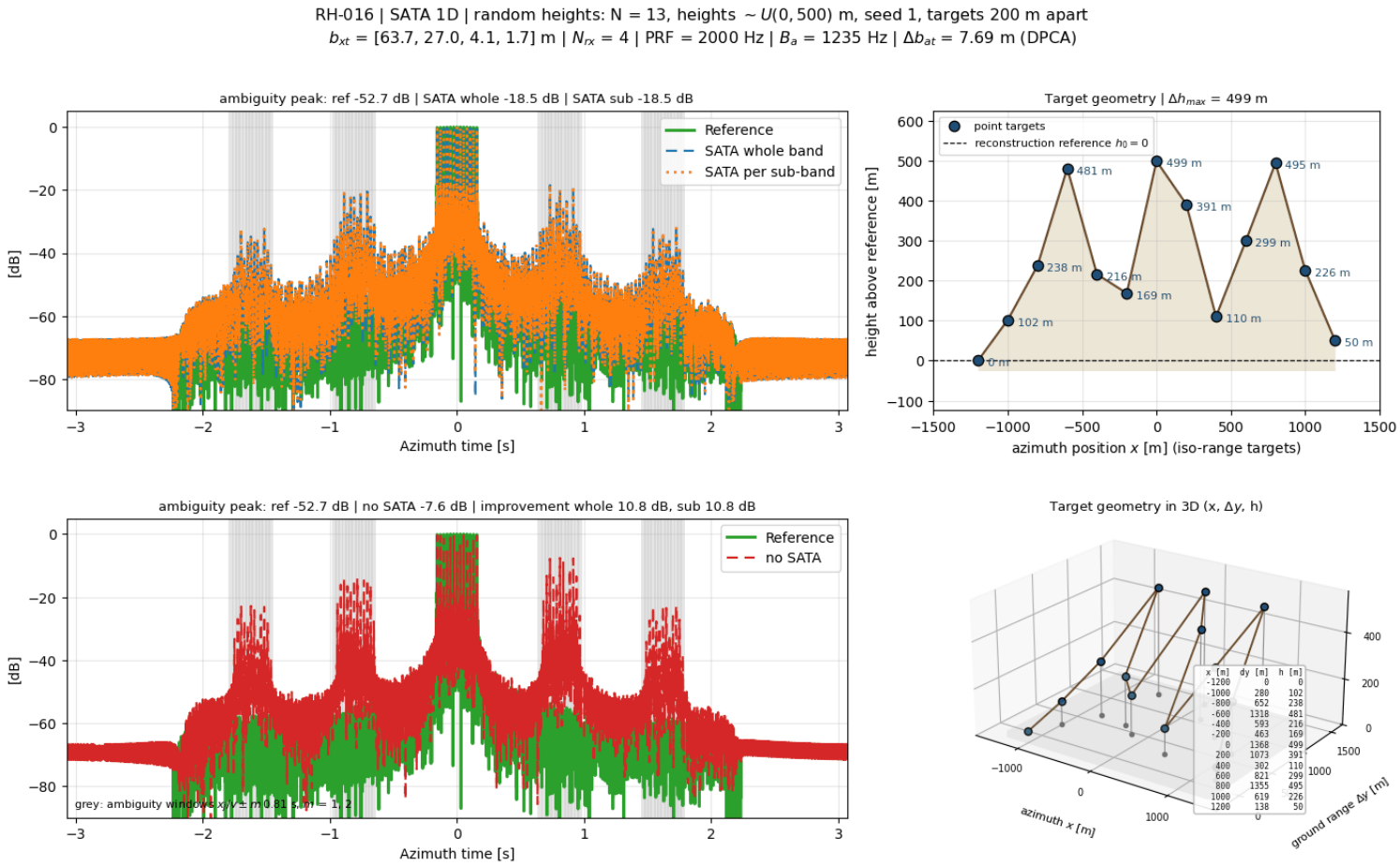


Figure 26: Random heights, 200 m apart: RH-016, $N = 13$, $h \sim U(0, 500)$ m, seed 1. Ambiguity peak: no SATA -7.6, SATA whole band -18.5, per sub-band -18.5 dB (improvement 10.8 dB).

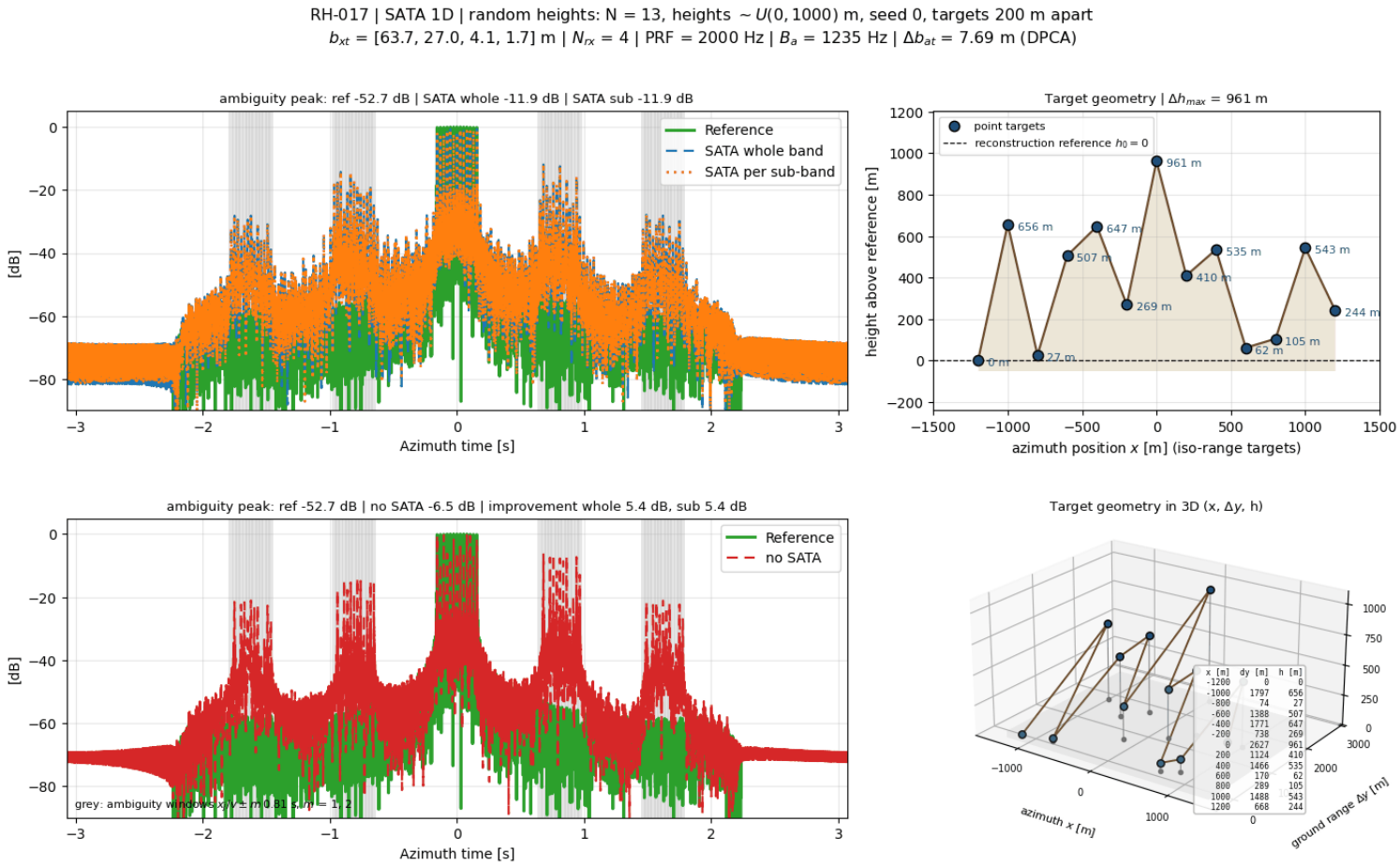


Figure 27: Random heights, 200 m apart: RH-017, $N = 13$, $h \sim U(0, 1000)$ m, seed 0. Ambiguity peak: no SATA -6.5, SATA whole band -11.9, per sub-band -11.9 dB (improvement 5.4 dB).

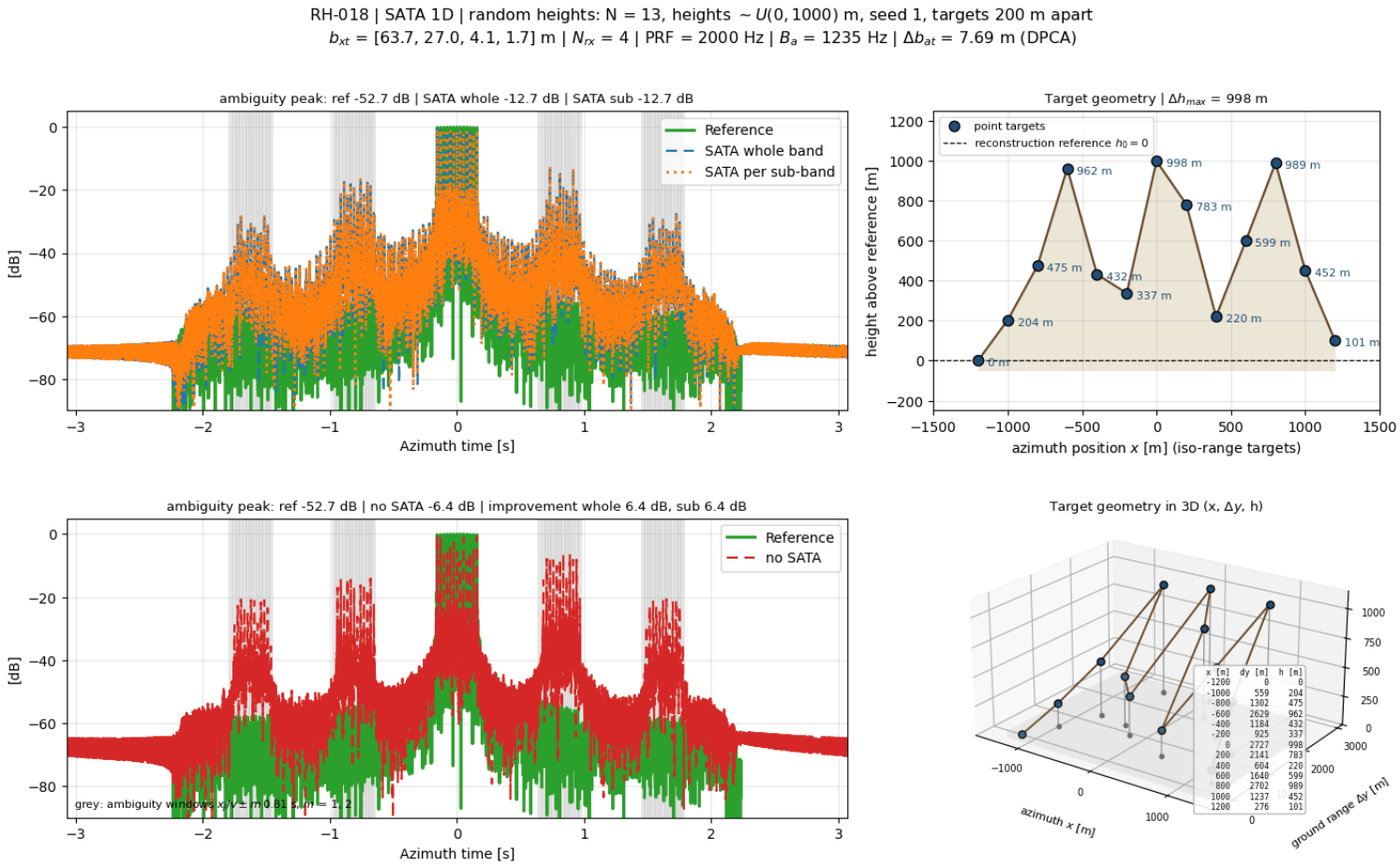


Figure 28: Random heights, 200 m apart: RH-018, $N = 13$, $h \sim U(0, 1000)$ m, seed 1. Ambiguity peak: no SATA −6.4, SATA whole band −12.7, per sub-band −12.7 dB (improvement 6.4 dB).

5 IRFs, random heights, targets 2.6 km apart (same heights)

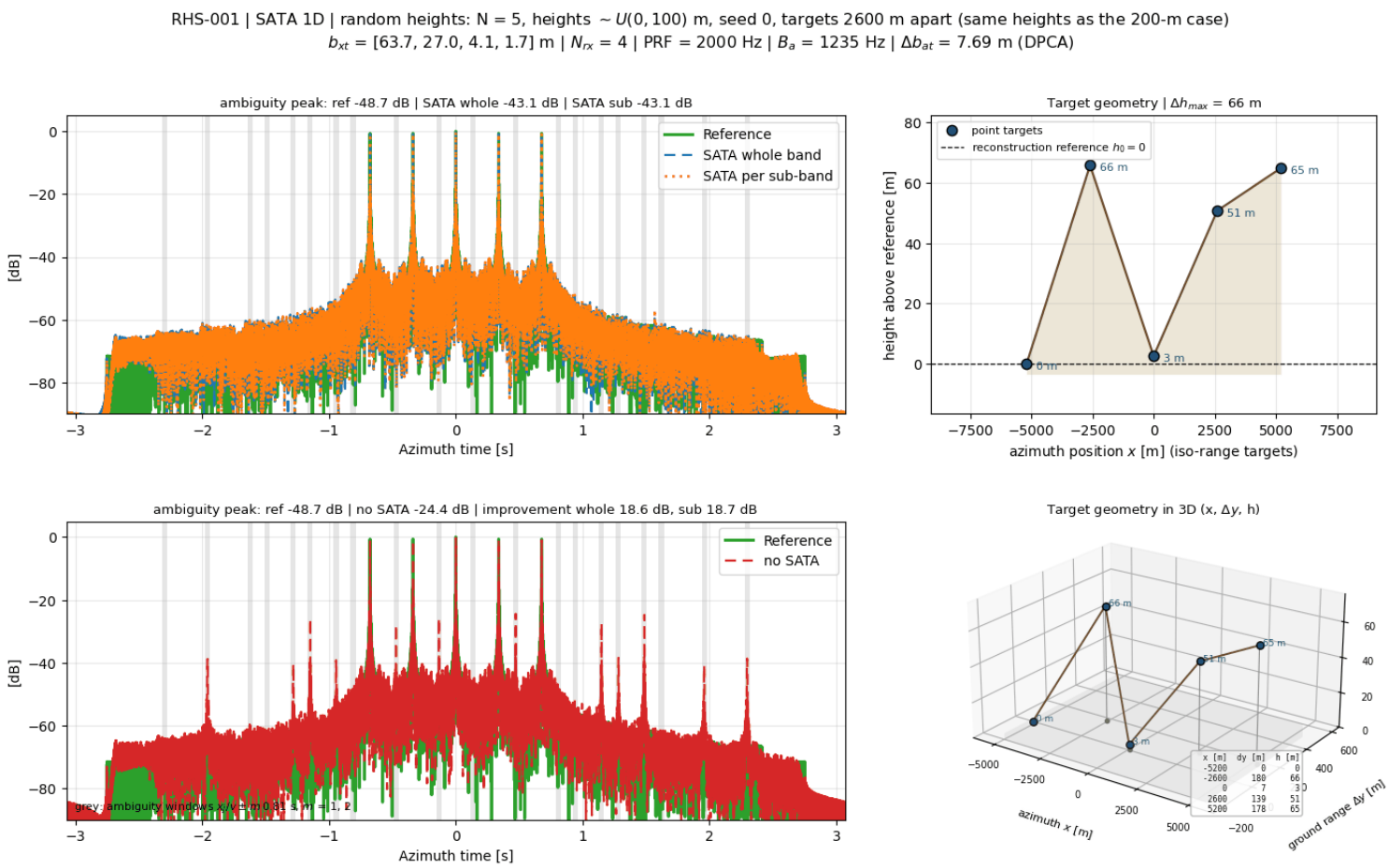


Figure 29: Random heights, 2.6 km apart, same heights: RHS-001, $N = 5$, $h \sim U(0, 100)$ m, seed 0. Ambiguity peak: no SATA -24.4, SATA whole band -43.1, per sub-band -43.1 dB (improvement 18.6 dB).

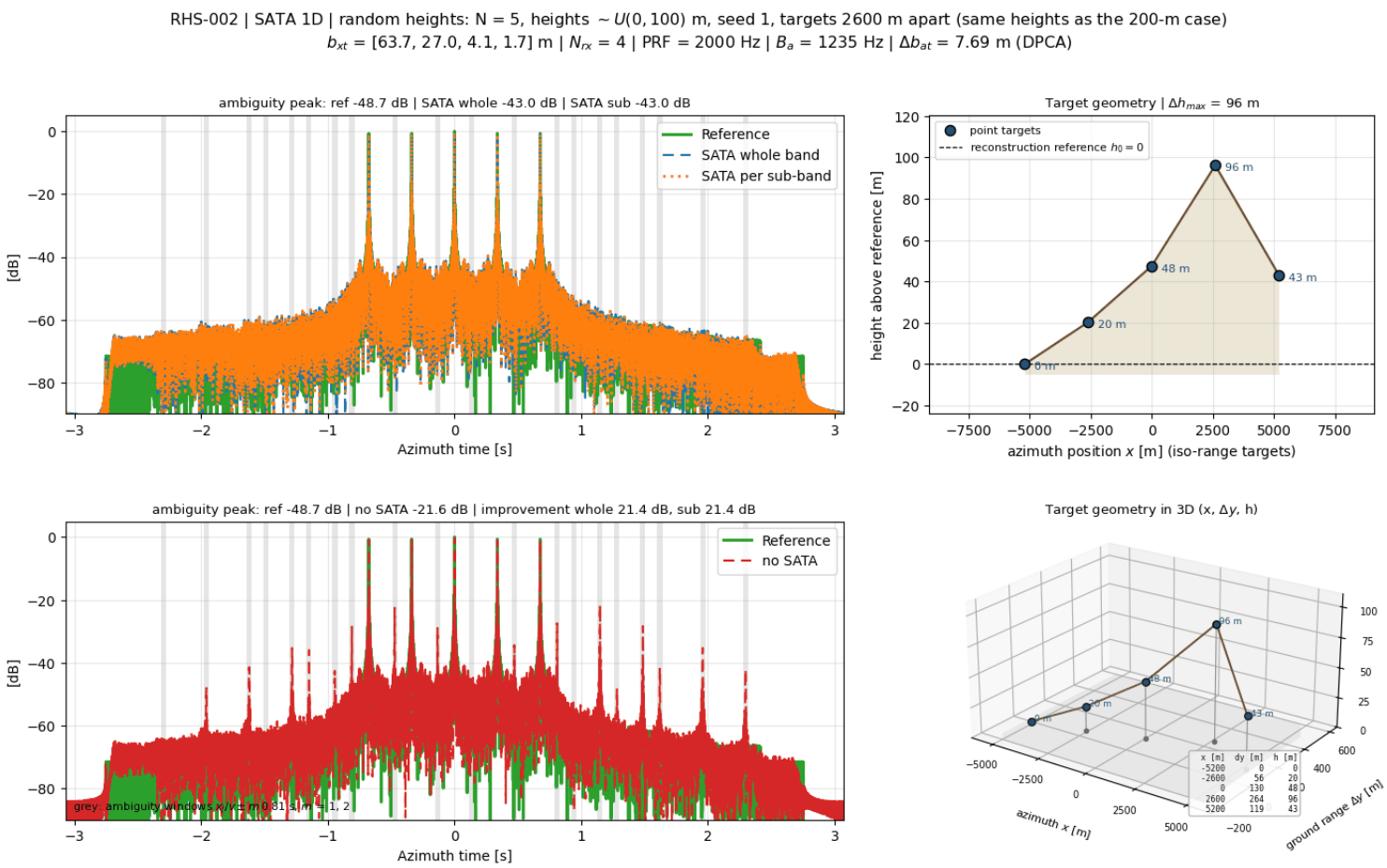


Figure 30: Random heights, 2.6 km apart, same heights: RHS-002, $N = 5$, $h \sim U(0, 100)$ m, seed 1. Ambiguity peak: no SATA −21.6, SATA whole band −43.0, per sub-band −43.0 dB (improvement 21.4 dB).

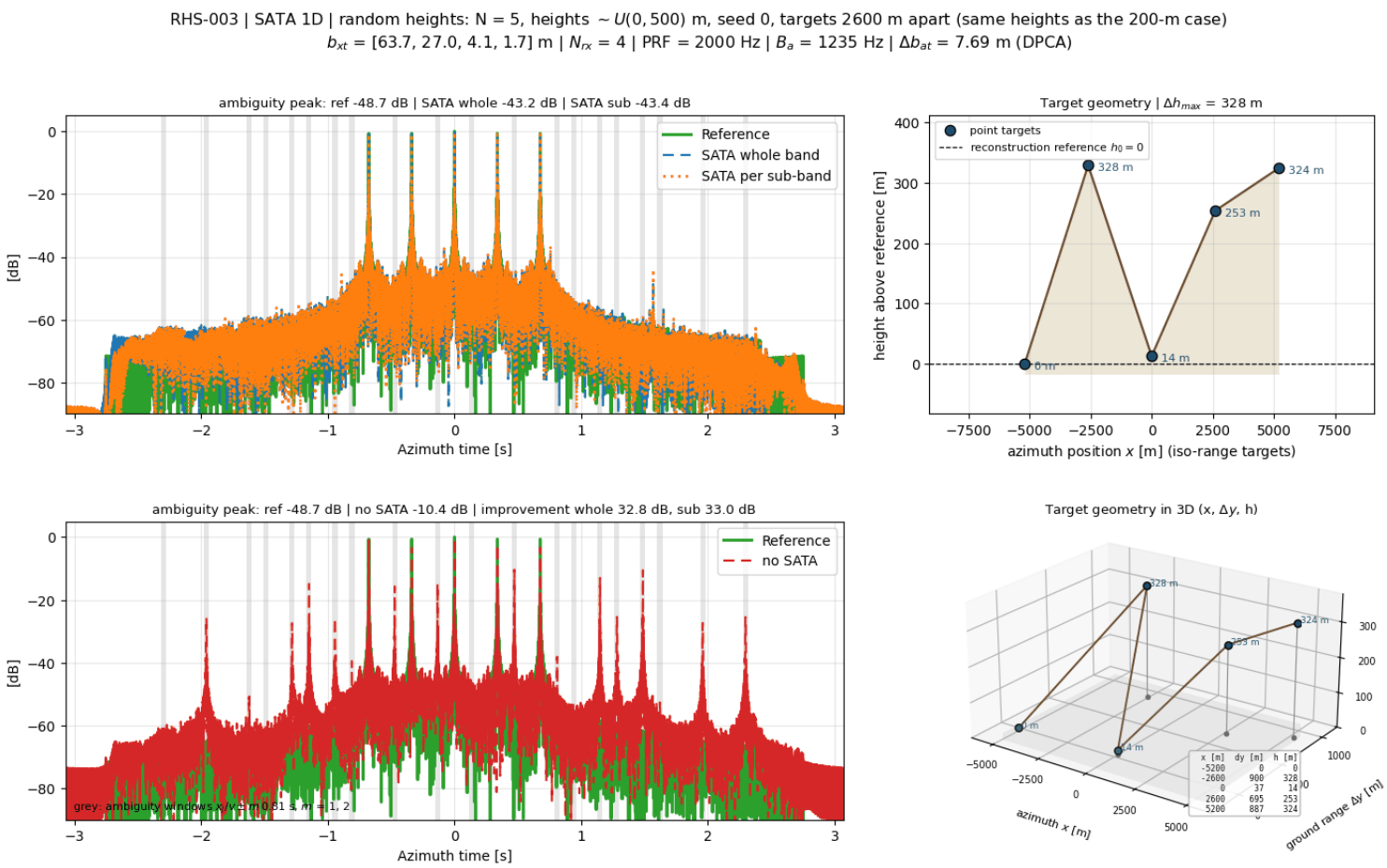


Figure 31: Random heights, 2.6 km apart, same heights: RHS-003, $N = 5$, $h \sim U(0, 500)$ m, seed 0. Ambiguity peak: no SATA -10.4, SATA whole band -43.2, per sub-band -43.4 dB (improvement 32.8 dB).

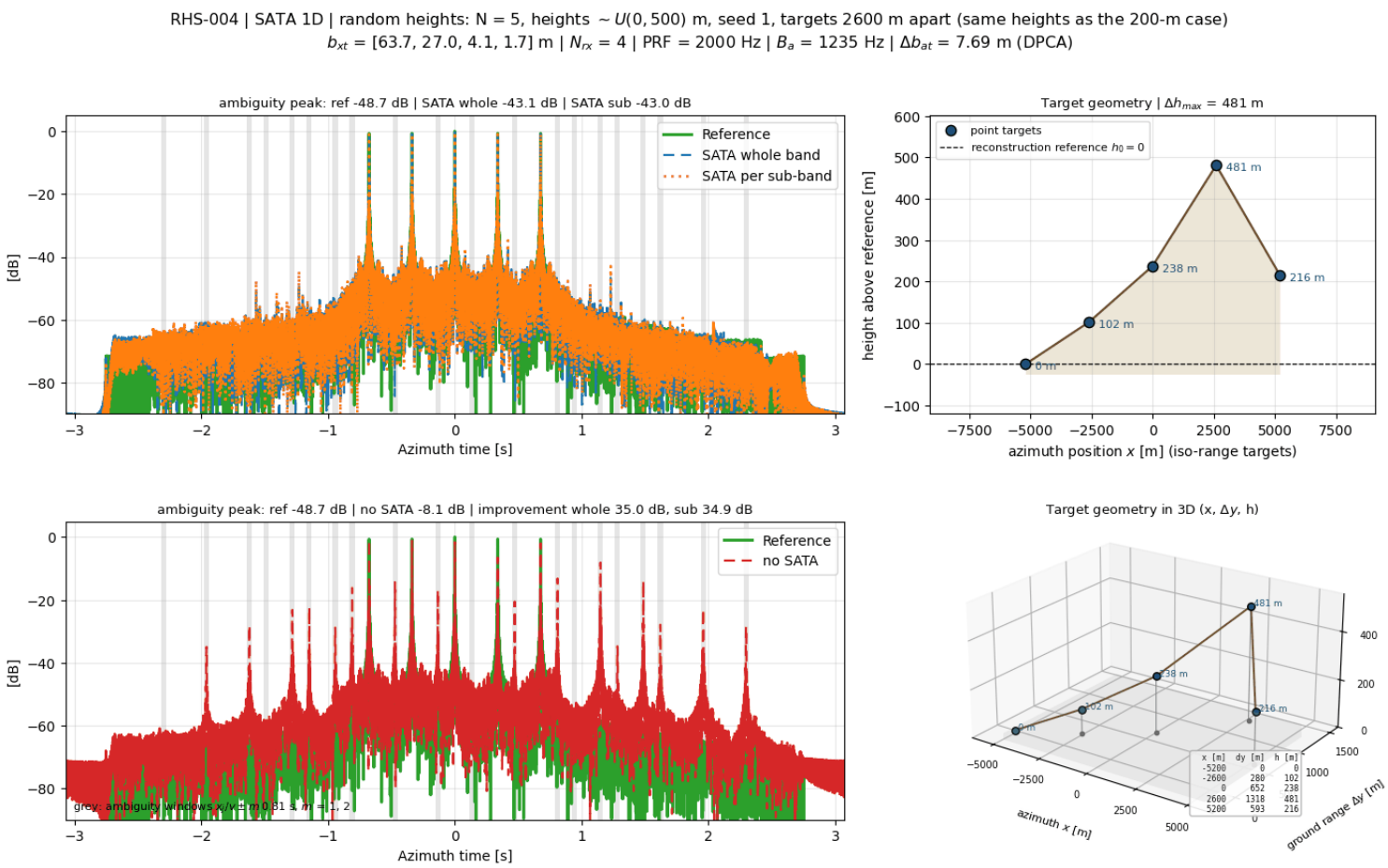


Figure 32: Random heights, 2.6 km apart, same heights: RHS-004, $N = 5$, $h \sim U(0, 500)$ m, seed 1. Ambiguity peak: no SATA -8.1, SATA whole band -43.1, per sub-band -43.0 dB (improvement 35.0 dB).

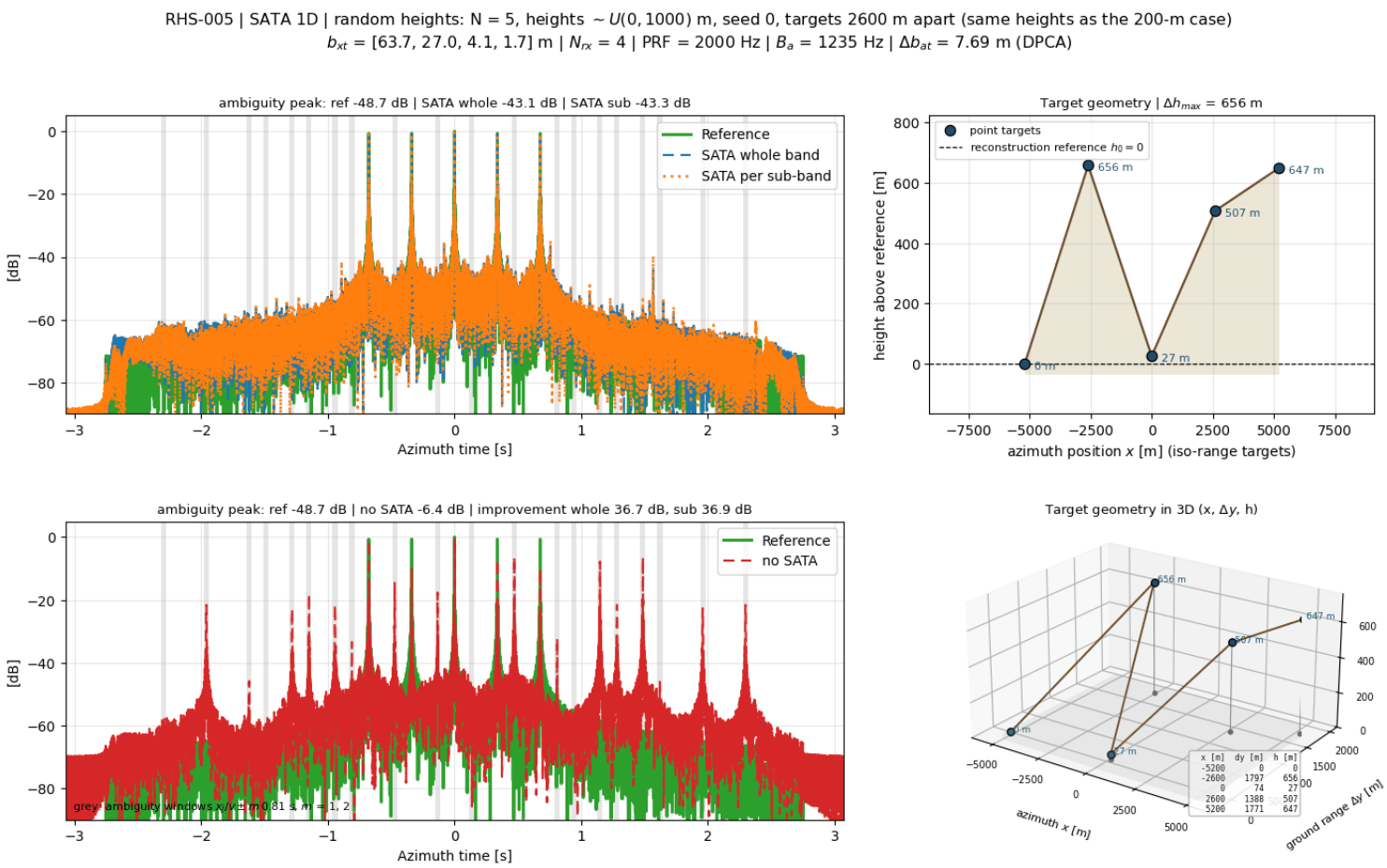


Figure 33: Random heights, 2.6 km apart, same heights: RHS-005, $N = 5$, $h \sim U(0, 1000)$ m, seed 0. Ambiguity peak: no SATA -6.4, SATA whole band -43.1, per sub-band -43.3 dB (improvement 36.7 dB).

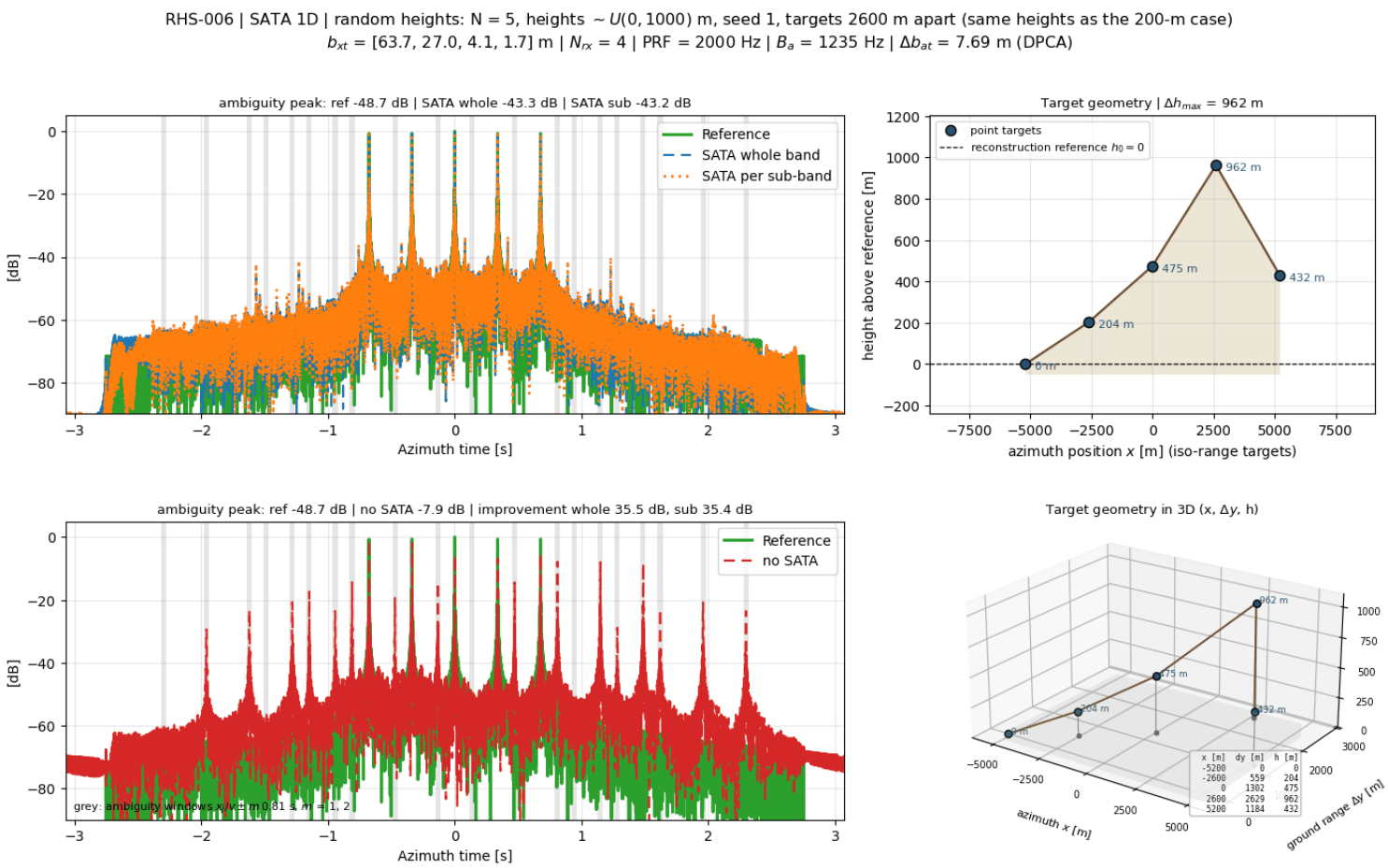


Figure 34: Random heights, 2.6 km apart, same heights: RHS-006, $N = 5$, $h \sim U(0, 1000)$ m, seed 1. Ambiguity peak: no SATA −7.9, SATA whole band −43.3, per sub-band −43.2 dB (improvement 35.5 dB).

6 IRFs, piecewise ramps, targets 200 m apart

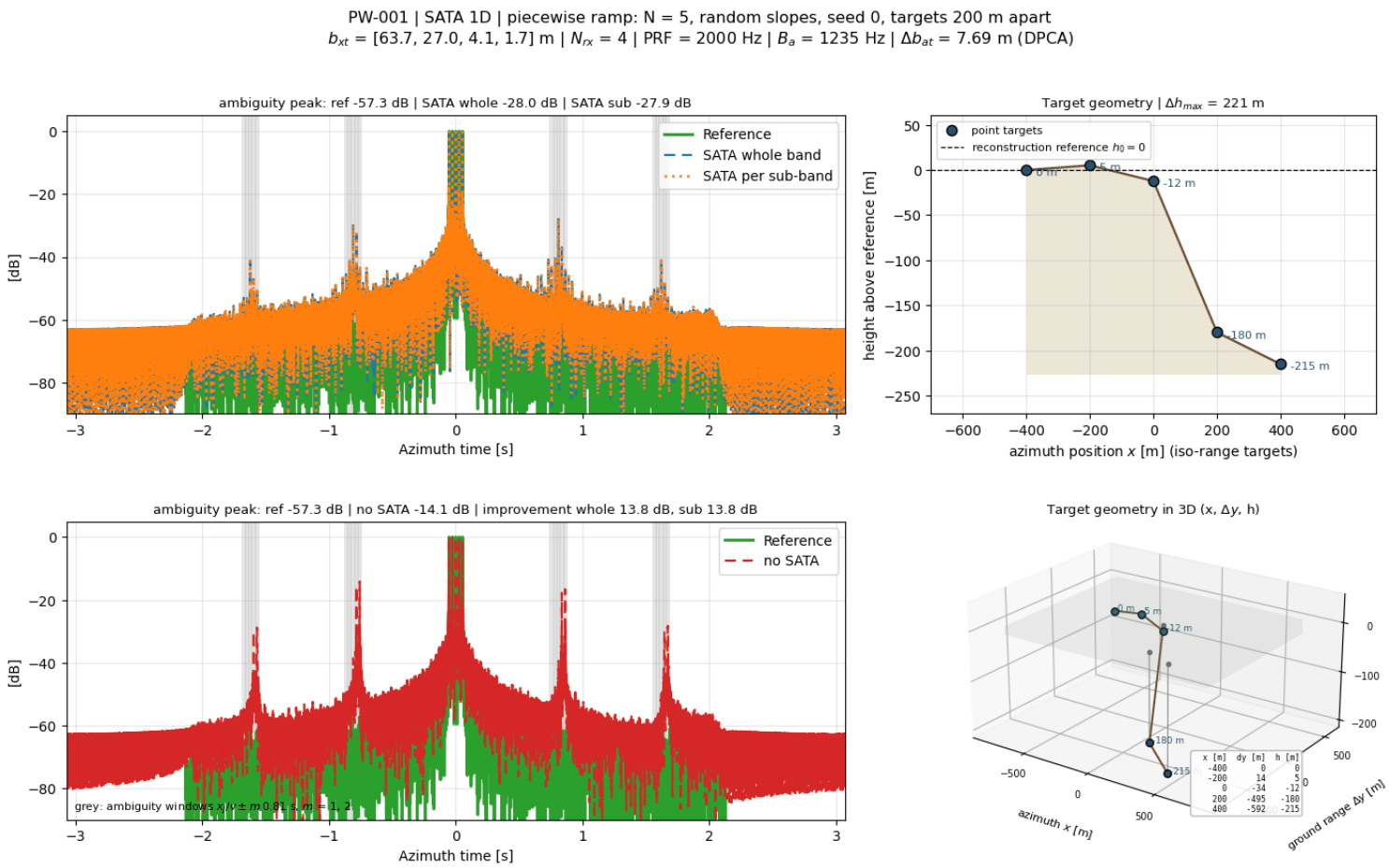


Figure 35: Piecewise ramp, 200 m apart: PW-001, $N = 5$, random slopes, seed 0. Ambiguity peak: no SATA −14.1, SATA whole band −28.0, per sub-band −27.9 dB (improvement 13.8 dB).

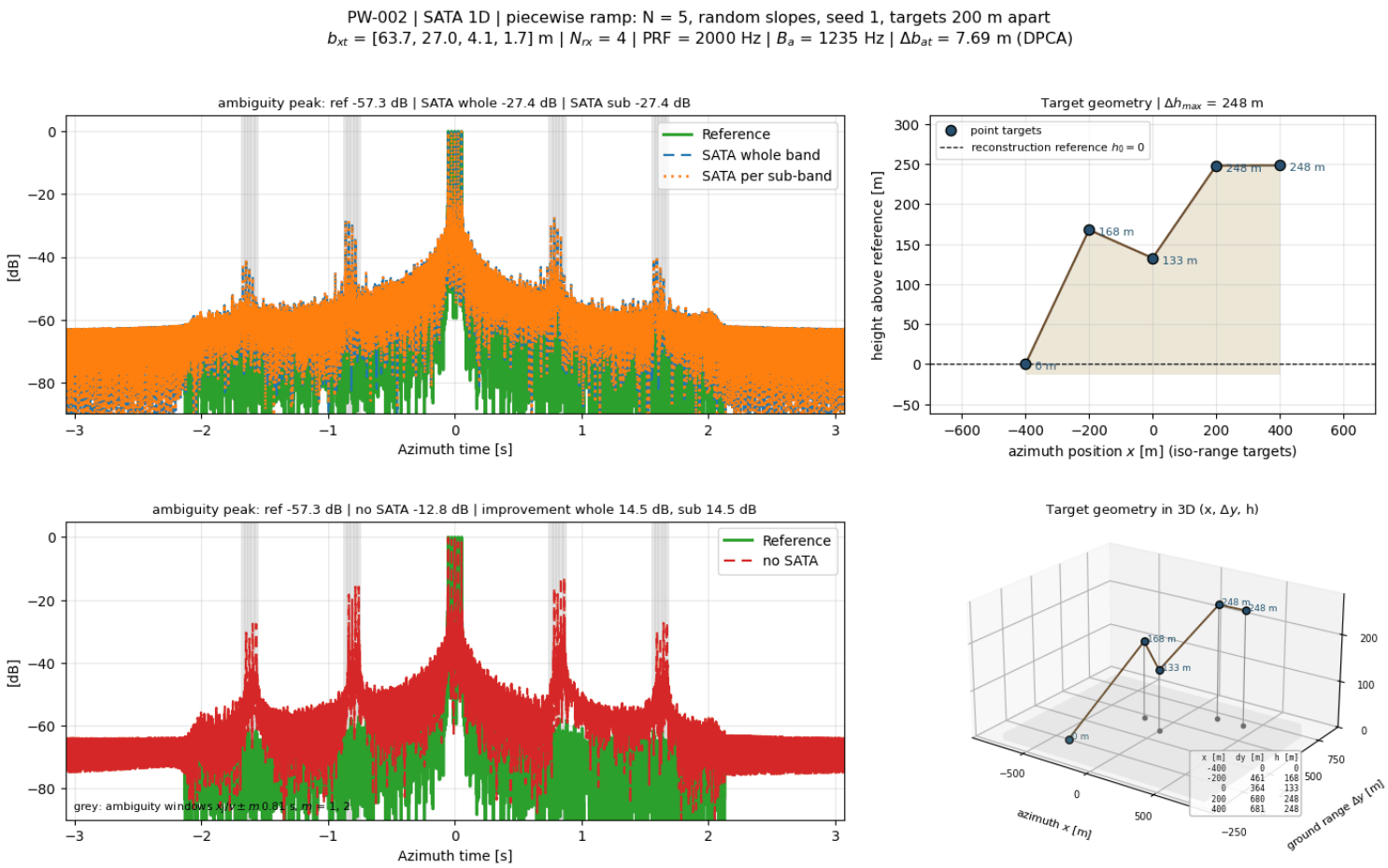


Figure 36: Piecewise ramp, 200 m apart: PW-002, $N = 5$, random slopes, seed 1. Ambiguity peak: no SATA −12.8, SATA whole band −27.4, per sub-band −27.4 dB (improvement 14.5 dB).

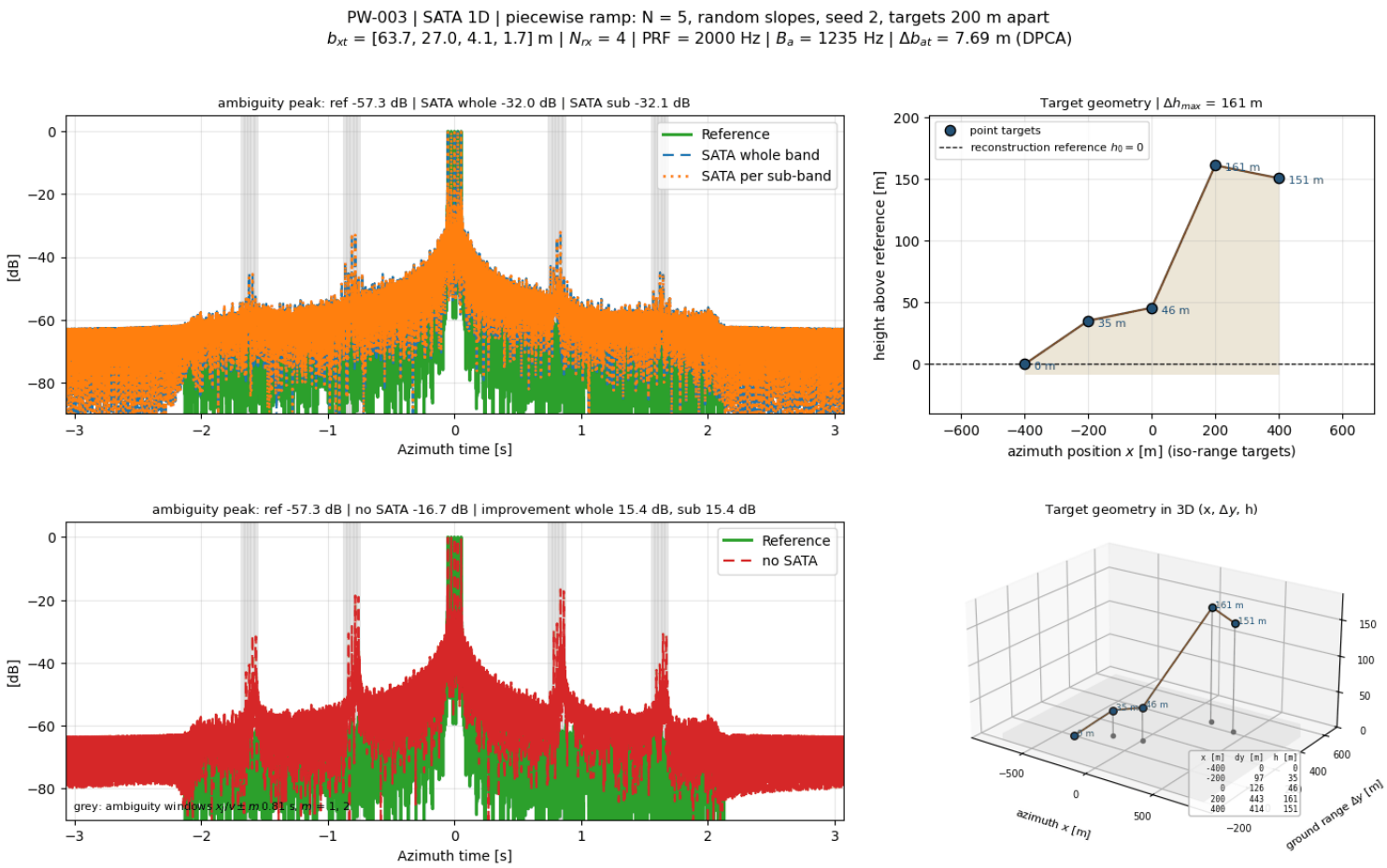


Figure 37: Piecewise ramp, 200 m apart: PW-003, $N = 5$, random slopes, seed 2. Ambiguity peak: no SATA −16.7, SATA whole band −32.0, per sub-band −32.1 dB (improvement 15.4 dB).

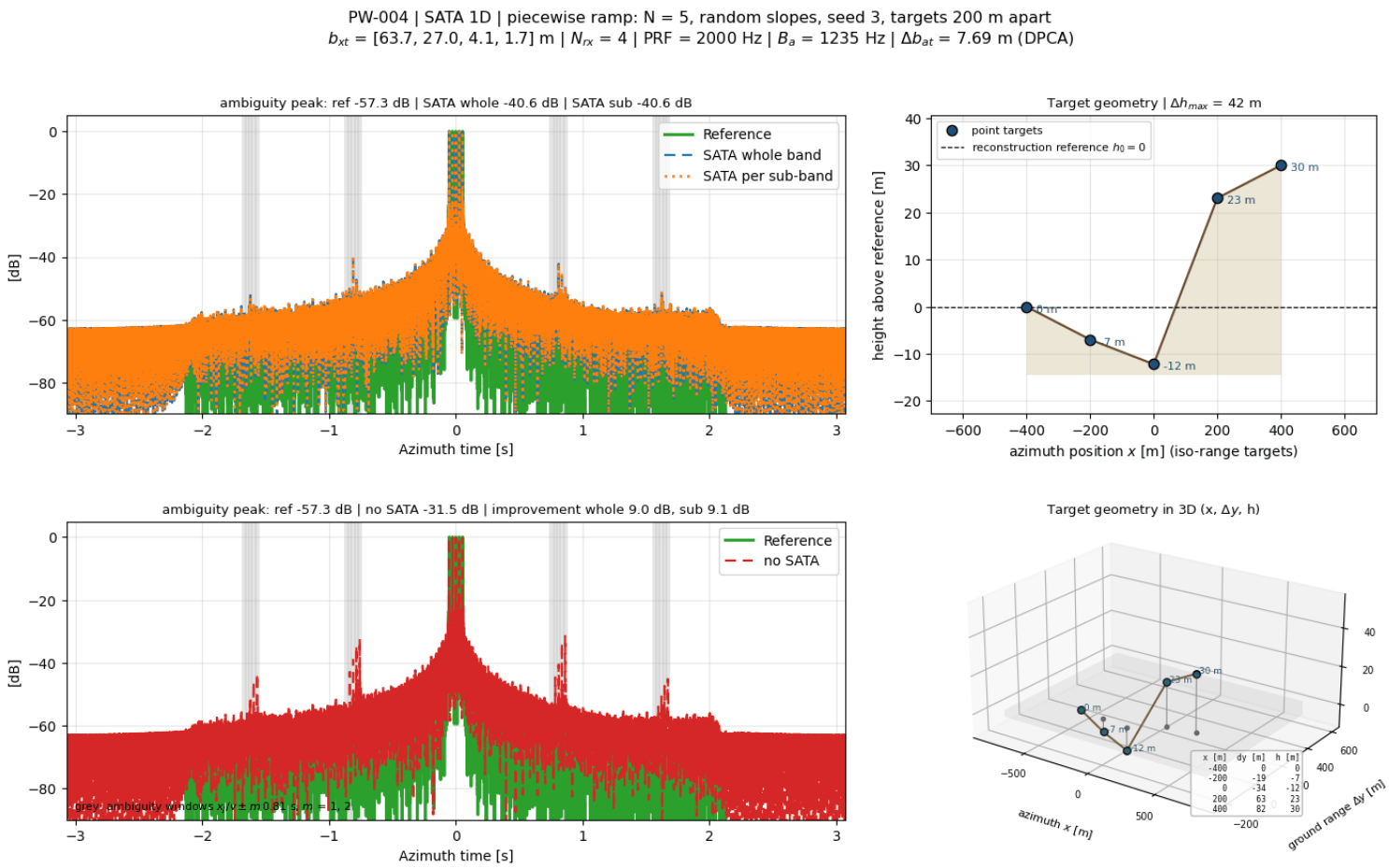


Figure 38: Piecewise ramp, 200 m apart: PW-004, $N = 5$, random slopes, seed 3. Ambiguity peak: no SATA −31.5, SATA whole band −40.6, per sub-band −40.6 dB (improvement 9.0 dB).

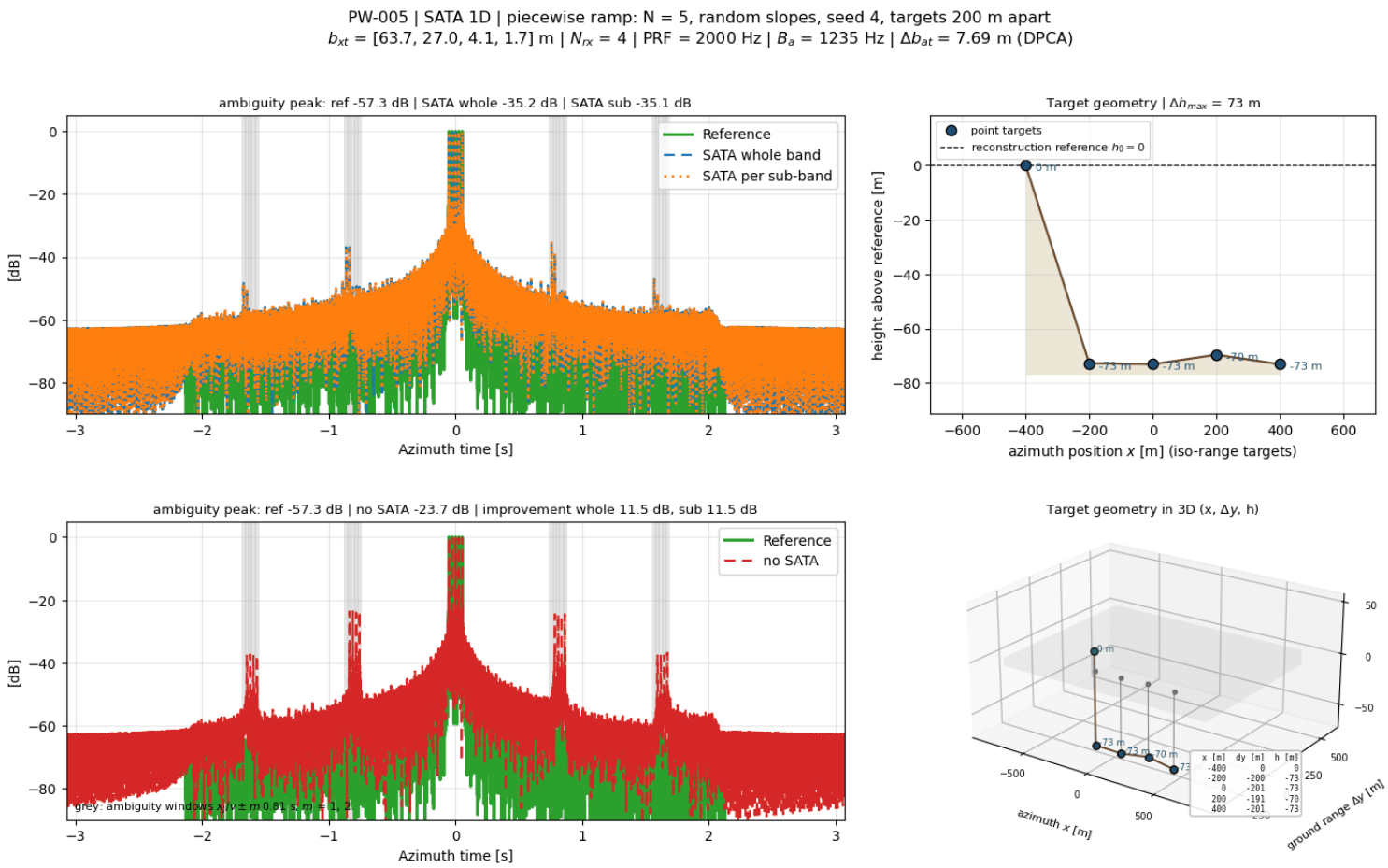


Figure 39: Piecewise ramp, 200 m apart: PW-005, $N = 5$, random slopes, seed 4. Ambiguity peak: no SATA −23.7, SATA whole band −35.2, per sub-band −35.1 dB (improvement 11.5 dB).

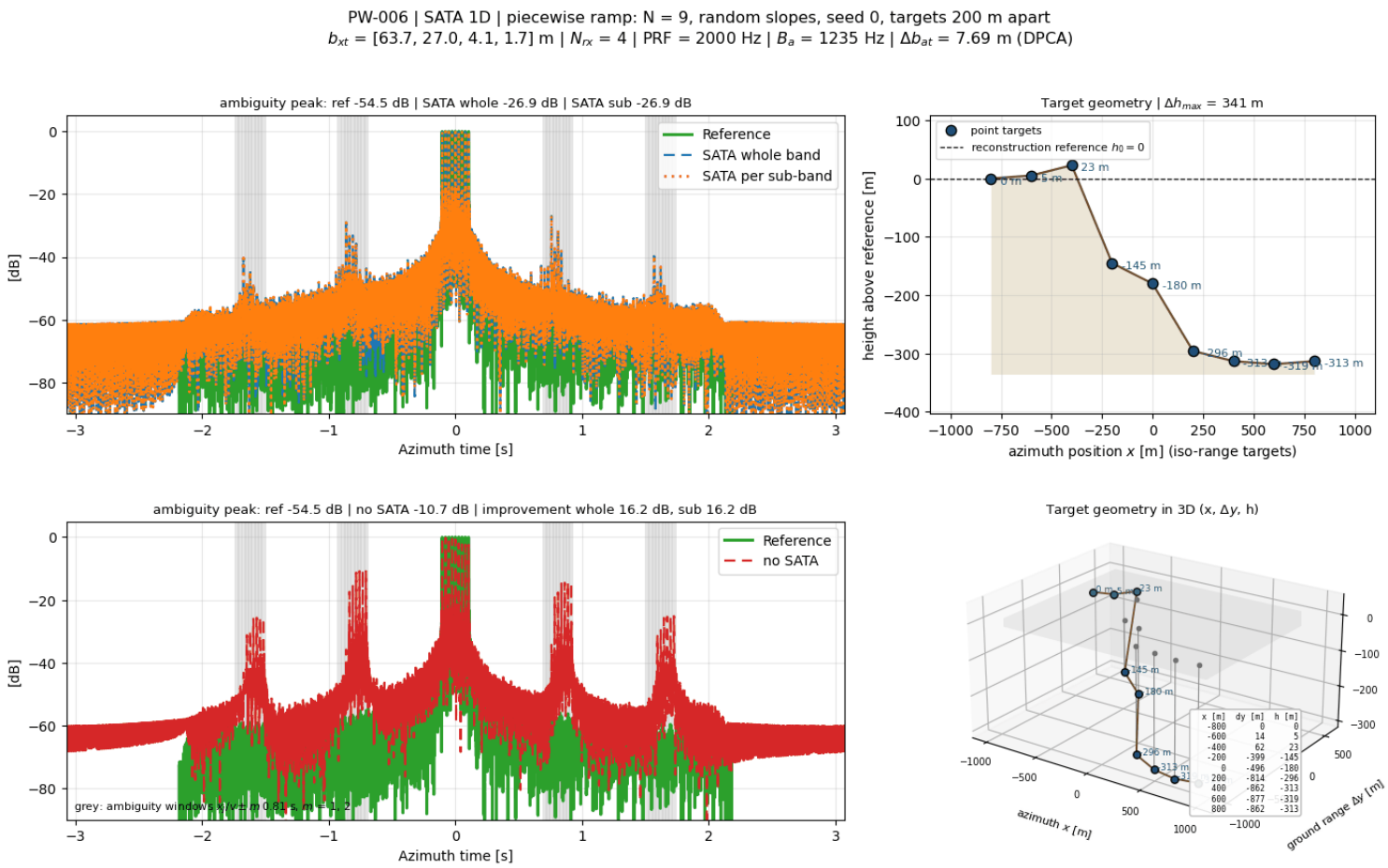


Figure 40: Piecewise ramp, 200 m apart: PW-006, $N = 9$, random slopes, seed 0. Ambiguity peak: no SATA −10.7, SATA whole band −26.9, per sub-band −26.9 dB (improvement 16.2 dB).

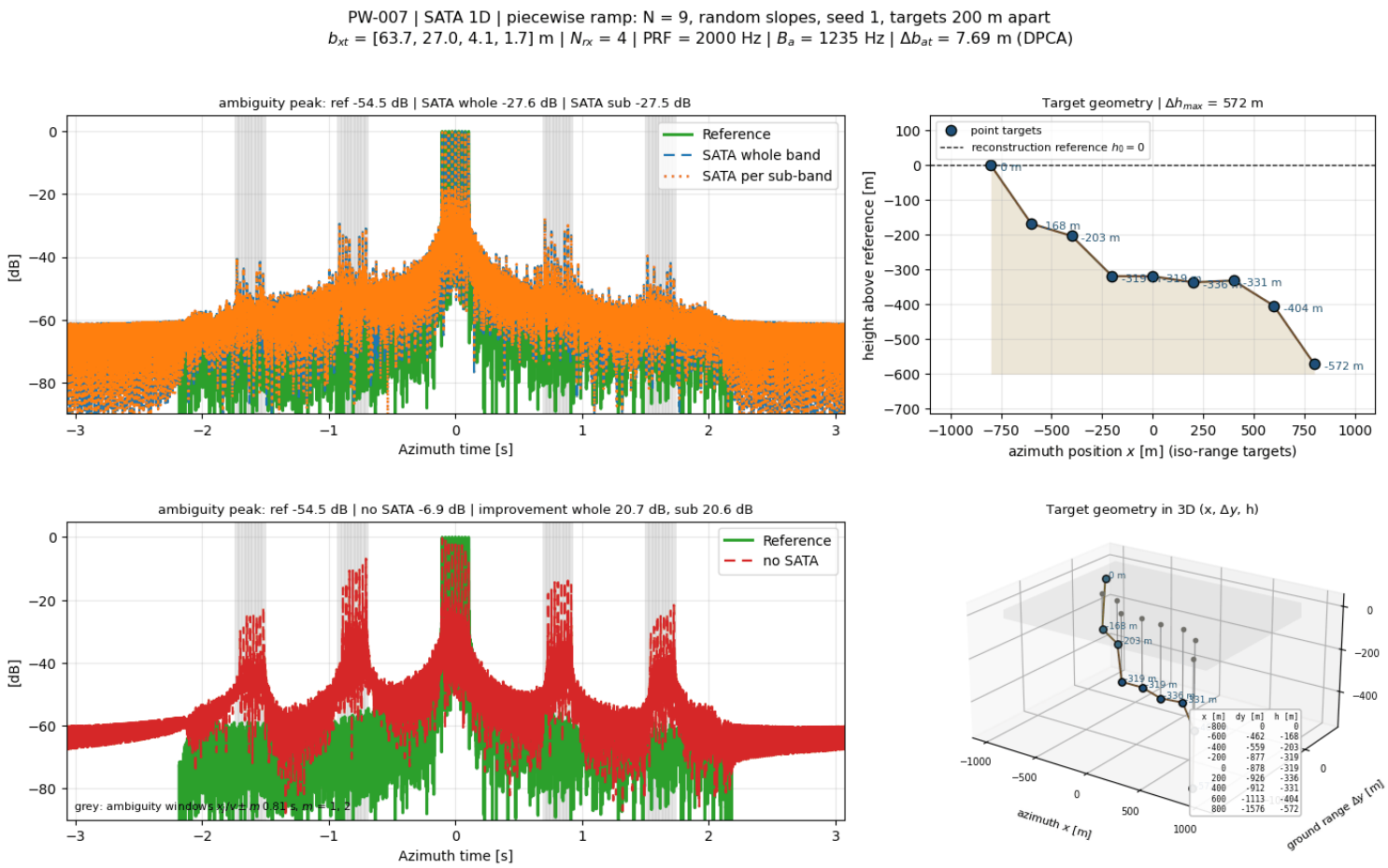


Figure 41: Piecewise ramp, 200 m apart: PW-007, $N = 9$, random slopes, seed 1. Ambiguity peak: no SATA −6.9, SATA whole band −27.6, per sub-band −27.5 dB (improvement 20.7 dB).

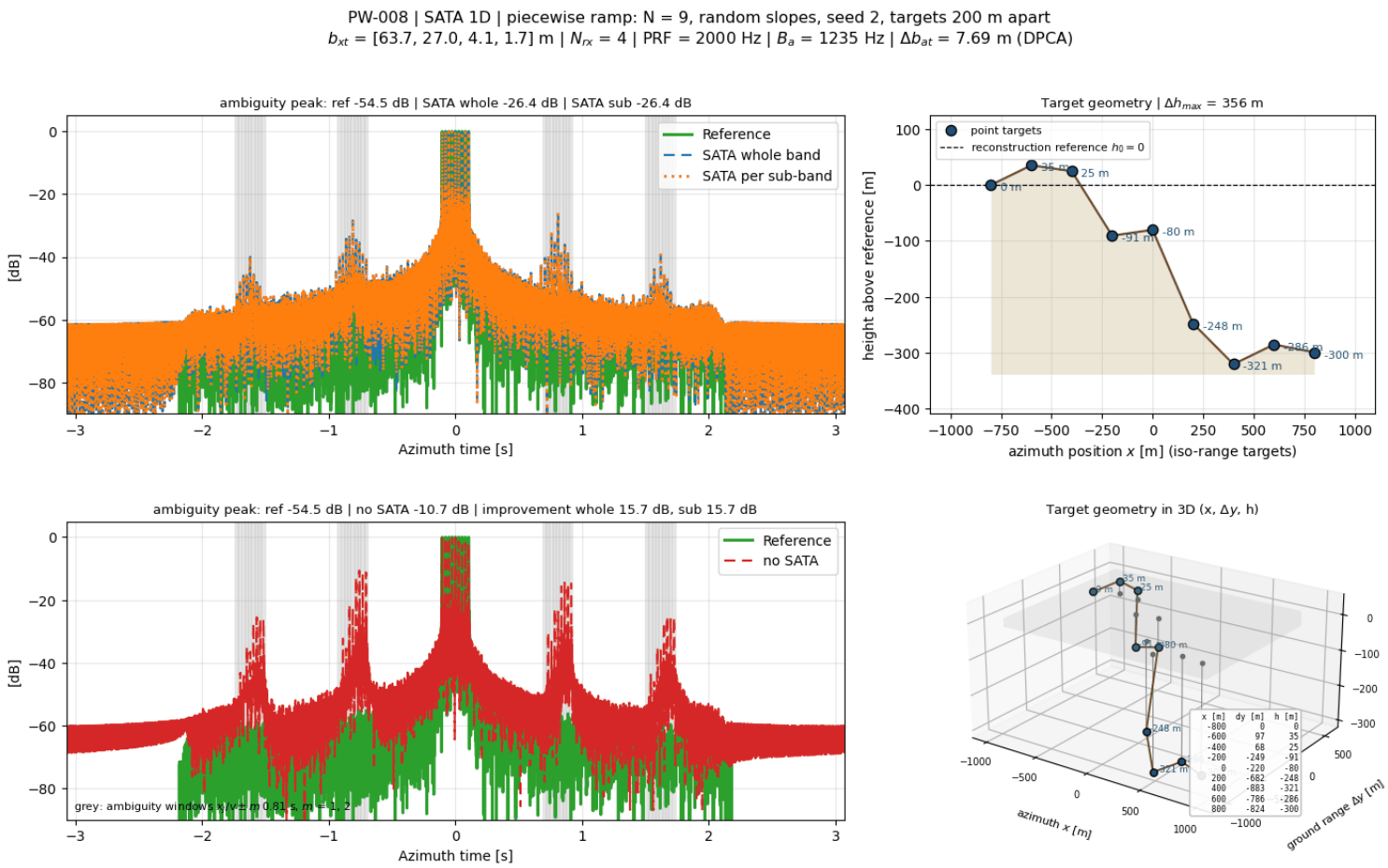


Figure 42: Piecewise ramp, 200 m apart: PW-008, $N = 9$, random slopes, seed 2. Ambiguity peak: no SATA −10.7, SATA whole band −26.4, per sub-band −26.4 dB (improvement 15.7 dB).

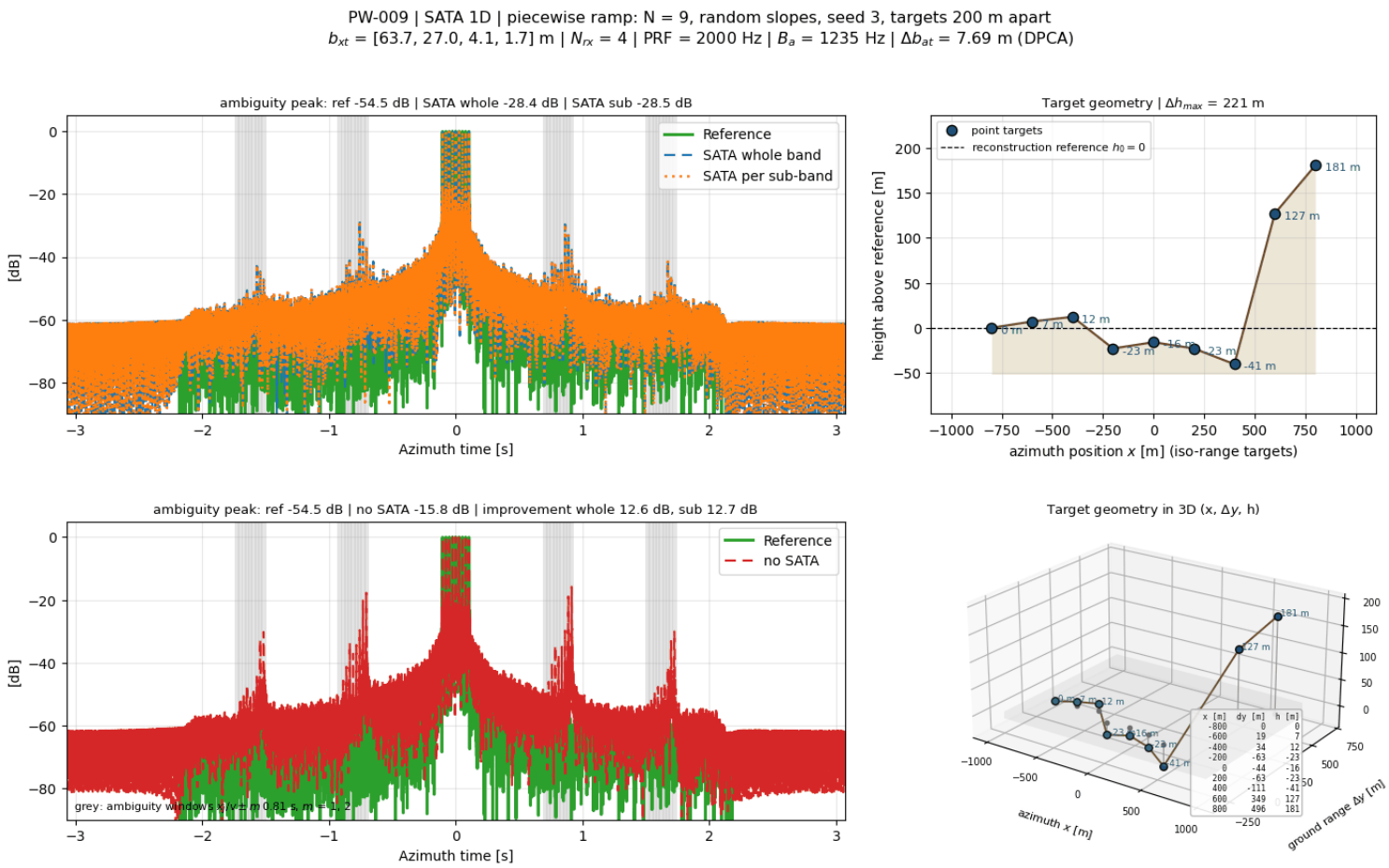


Figure 43: Piecewise ramp, 200 m apart: PW-009, $N = 9$, random slopes, seed 3. Ambiguity peak: no SATA −15.8, SATA whole band −28.4, per sub-band −28.5 dB (improvement 12.6 dB).

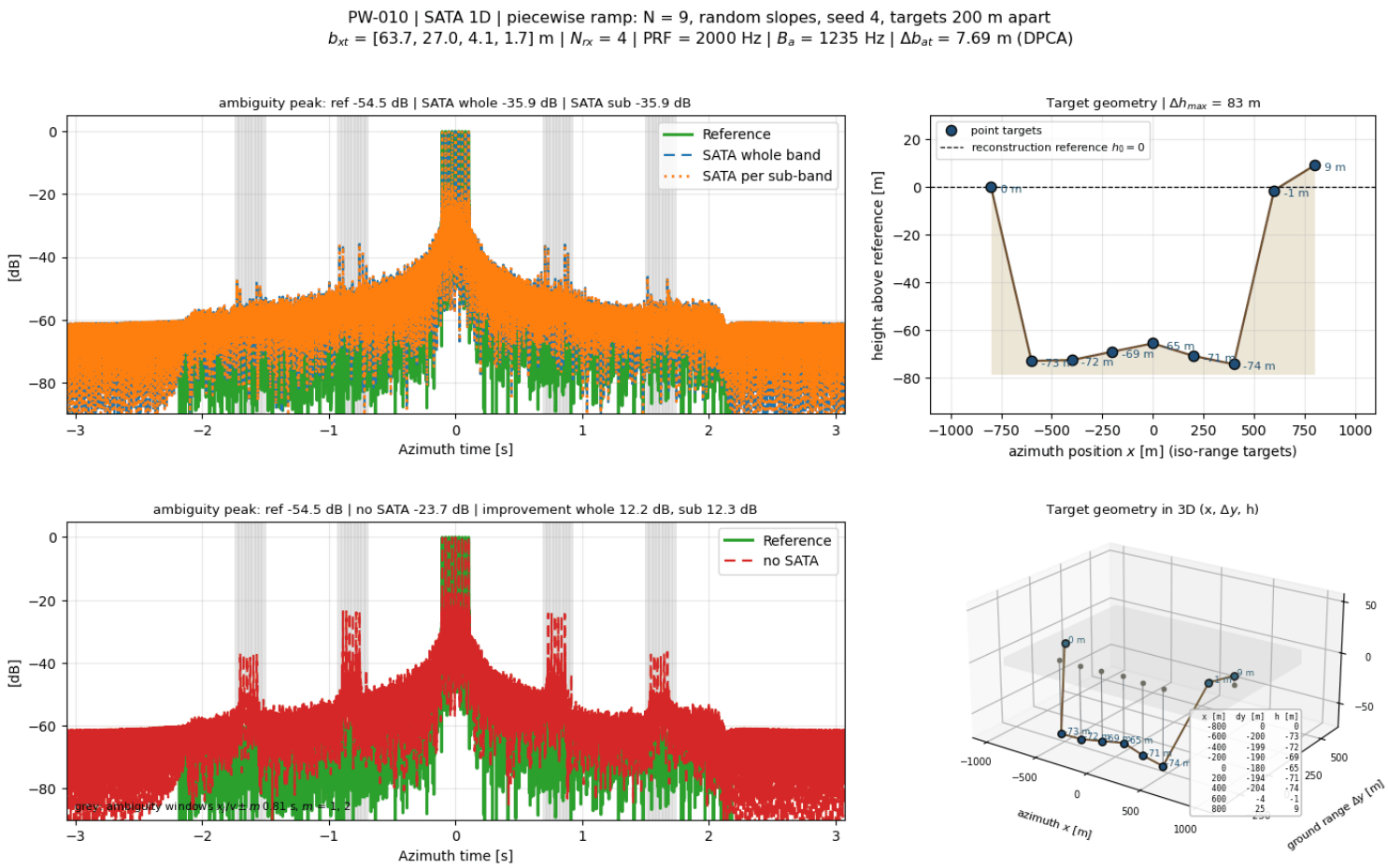


Figure 44: Piecewise ramp, 200 m apart: PW-010, $N = 9$, random slopes, seed 4. Ambiguity peak: no SATA −23.7, SATA whole band −35.9, per sub-band −35.9 dB (improvement 12.2 dB).

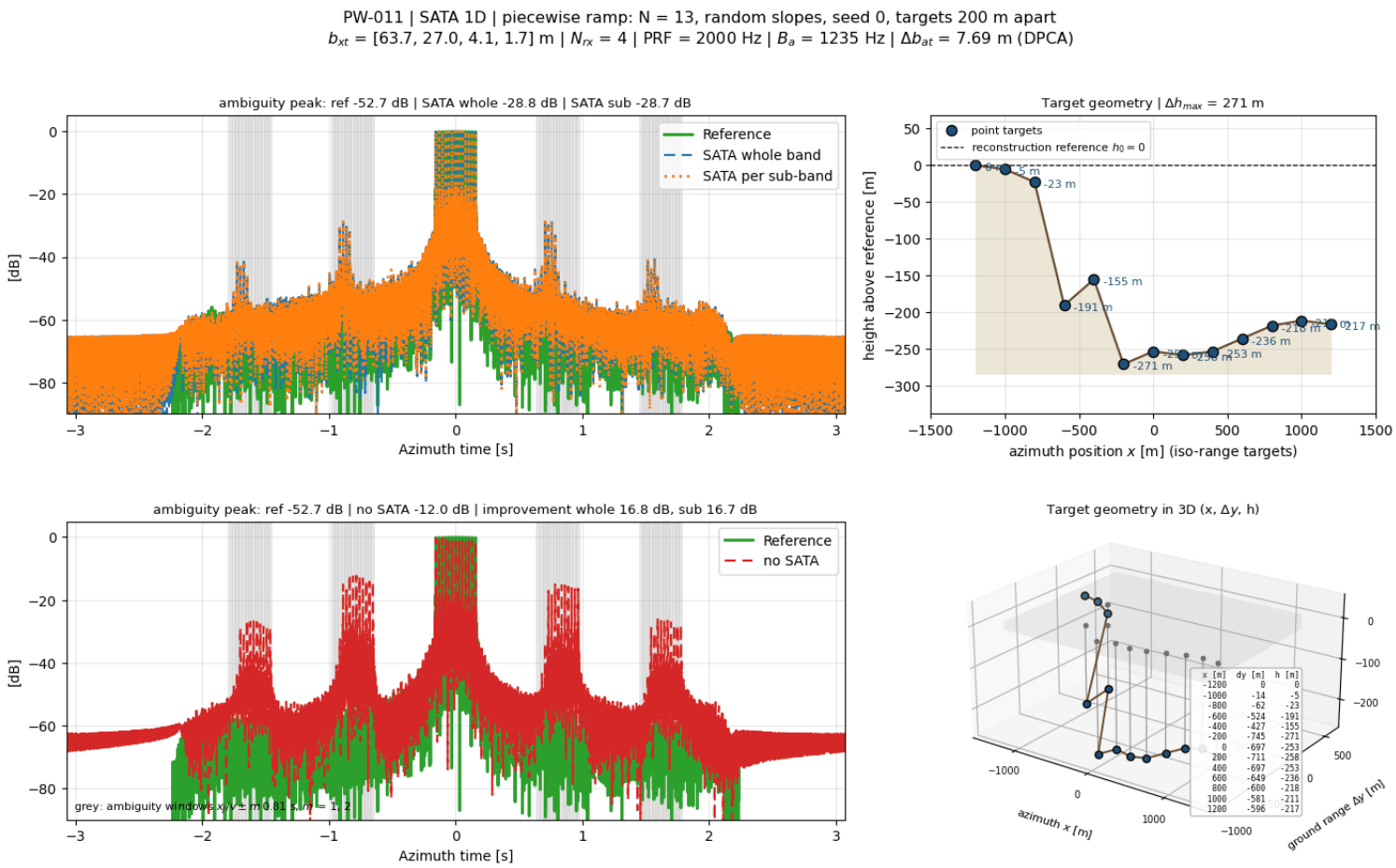


Figure 45: Piecewise ramp, 200 m apart: PW-011, $N = 13$, random slopes, seed 0. Ambiguity peak: no SATA −12.0, SATA whole band −28.8, per sub-band −28.7 dB (improvement 16.8 dB).

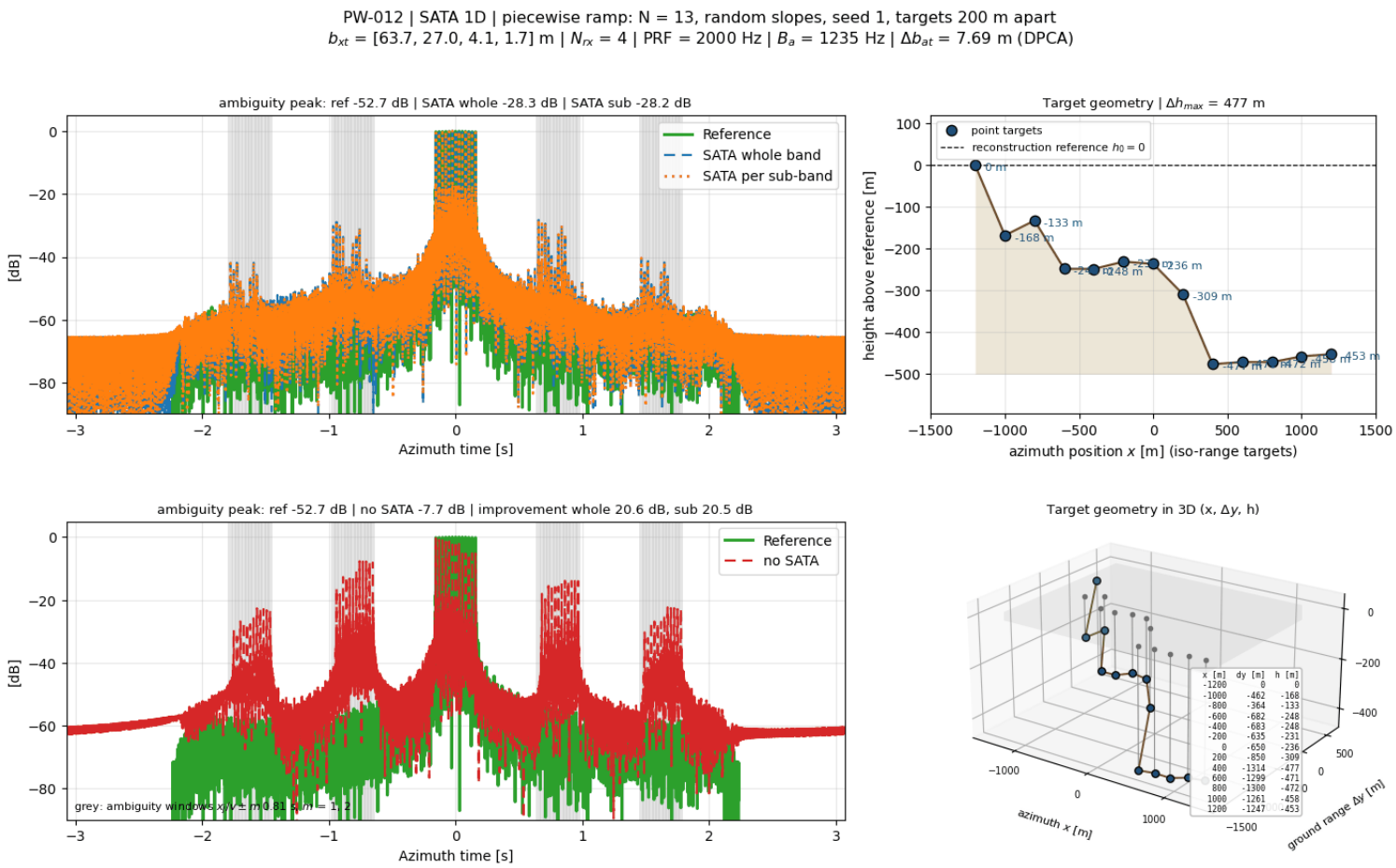


Figure 46: Piecewise ramp, 200 m apart: PW-012, $N = 13$, random slopes, seed 1. Ambiguity peak: no SATA −7.7, SATA whole band −28.3, per sub-band −28.2 dB (improvement 20.6 dB).

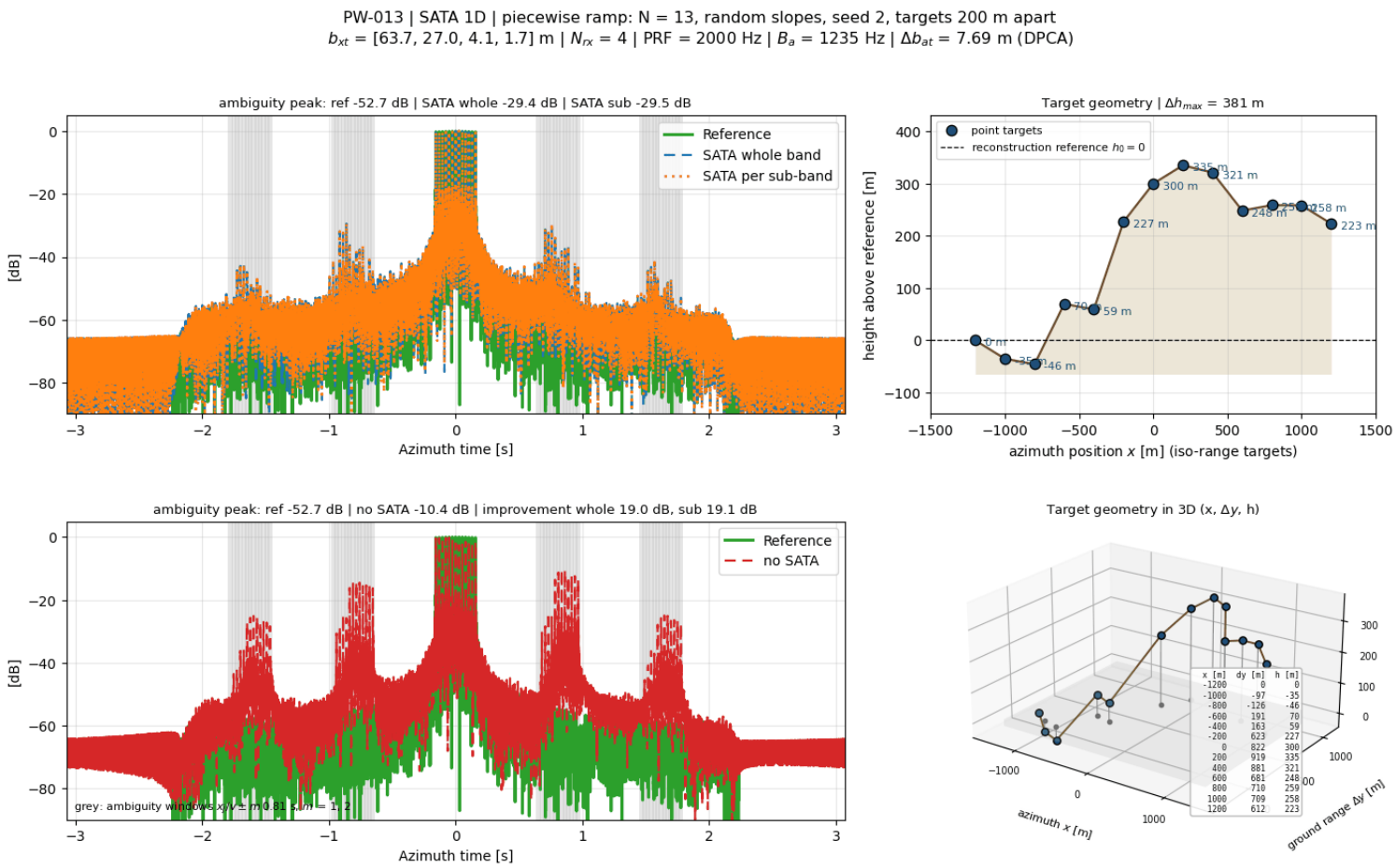


Figure 47: Piecewise ramp, 200 m apart: PW-013, $N = 13$, random slopes, seed 2. Ambiguity peak: no SATA −10.4, SATA whole band −29.4, per sub-band −29.5 dB (improvement 19.0 dB).

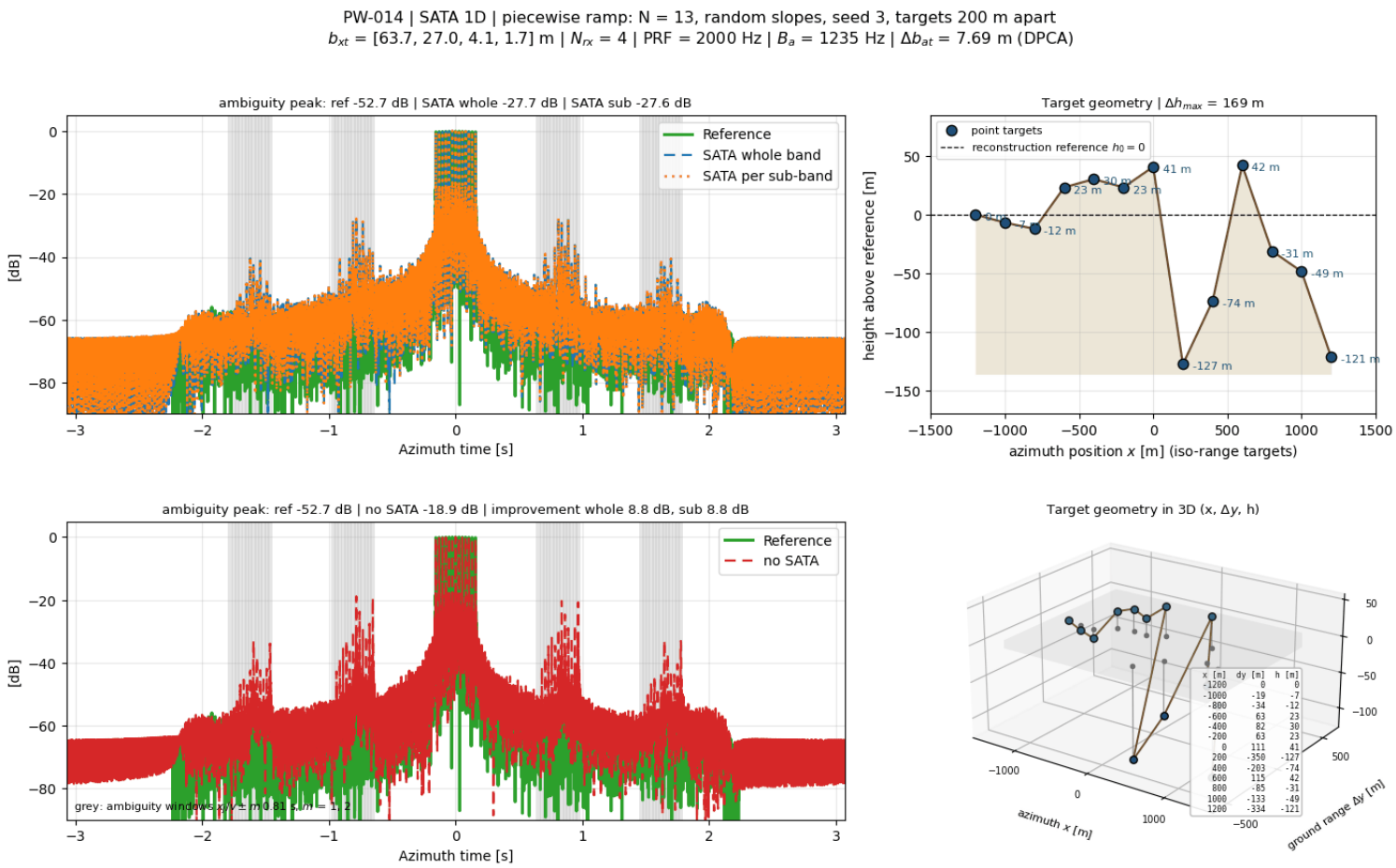


Figure 48: Piecewise ramp, 200 m apart: PW-014, $N = 13$, random slopes, seed 3. Ambiguity peak: no SATA −18.9, SATA whole band −27.7, per sub-band −27.6 dB (improvement 8.8 dB).

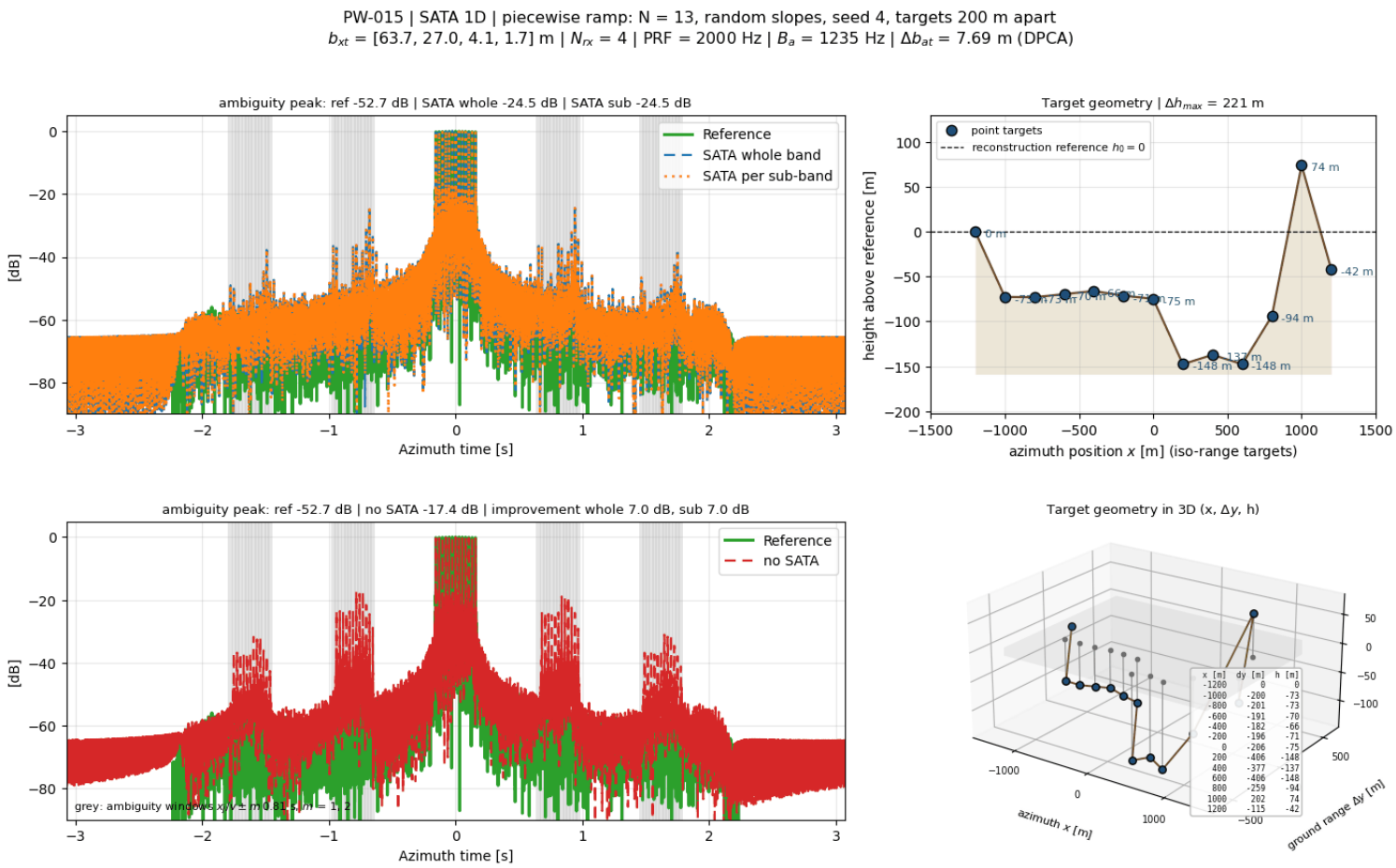


Figure 49: Piecewise ramp, 200 m apart: PW-015, $N = 13$, random slopes, seed 4. Ambiguity peak: no SATA −17.4, SATA whole band −24.5, per sub-band −24.5 dB (improvement 7.0 dB).

7 IRFs, piecewise ramps, targets 2.6 km apart (same heights)

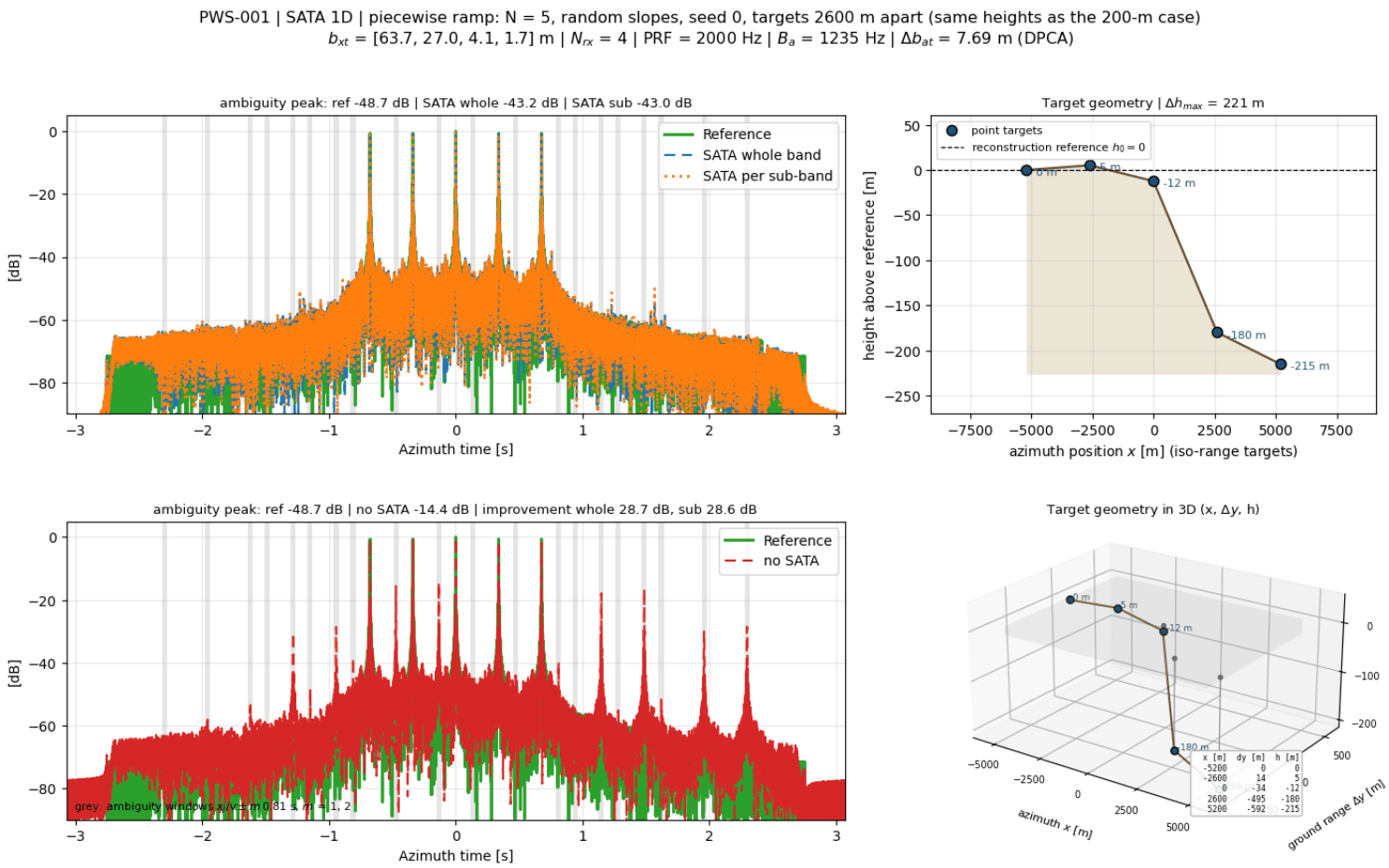


Figure 50: Piecewise ramp, 2.6 km apart, same heights: PWS-001, $N = 5$, random slopes, seed 0. Ambiguity peak: no SATA −14.4, SATA whole band −43.2, per sub-band −43.0 dB (improvement 28.7 dB).

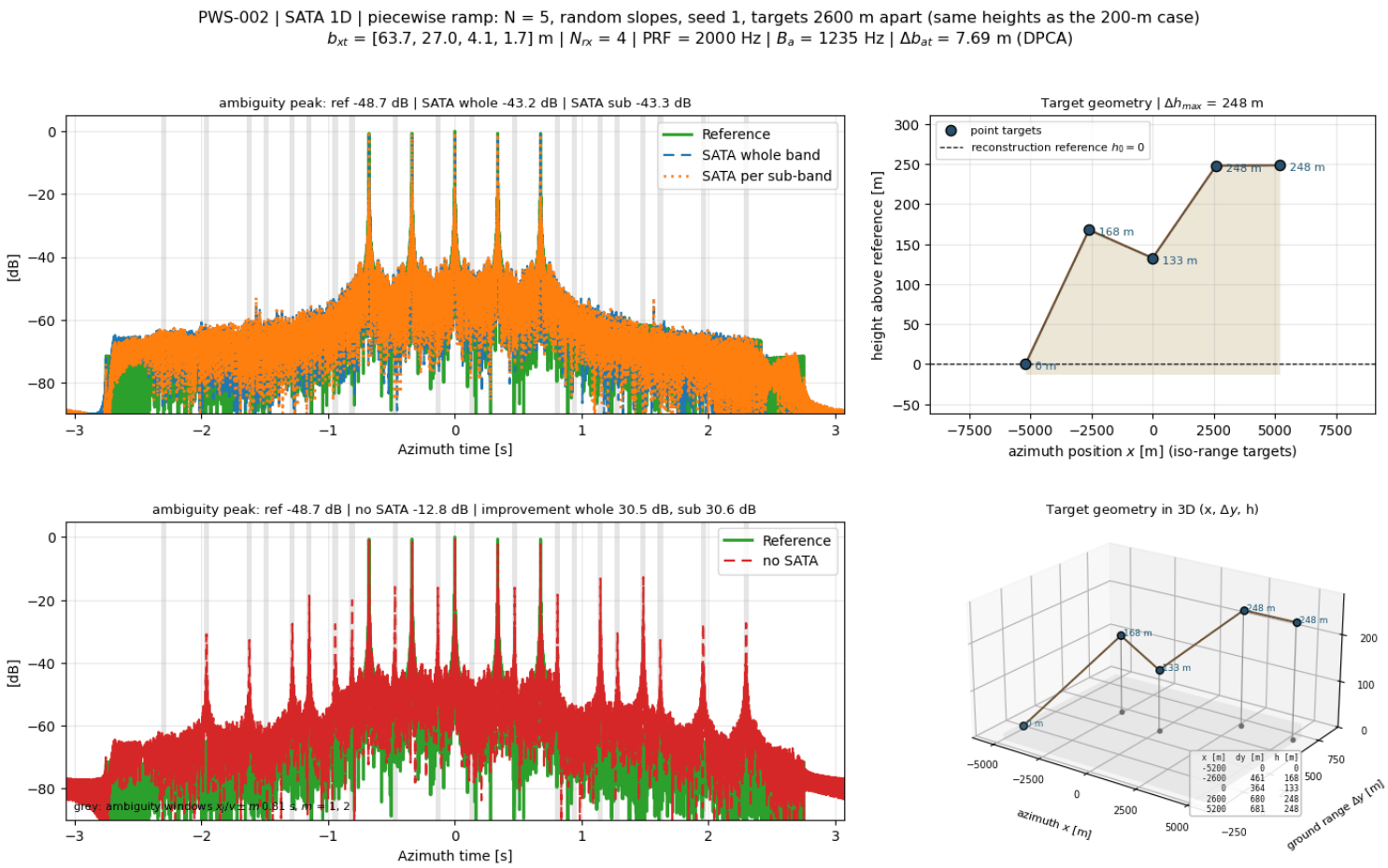


Figure 51: Piecewise ramp, 2.6 km apart, same heights: PWS-002, $N = 5$, random slopes, seed 1. Ambiguity peak: no SATA −12.8, SATA whole band −43.2, per sub-band −43.3 dB (improvement 30.5 dB).

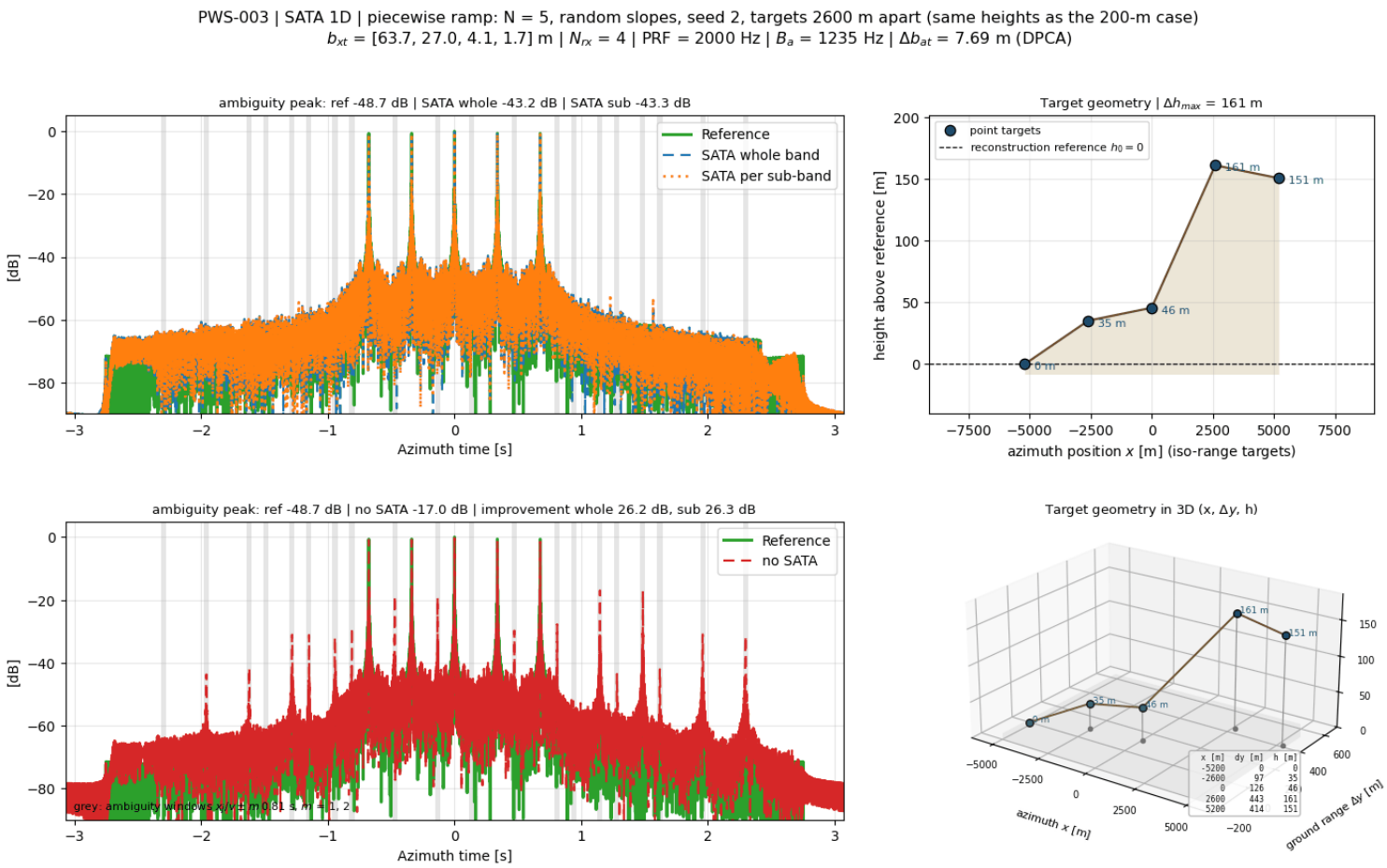


Figure 52: Piecewise ramp, 2.6 km apart, same heights: PWS-003, $N = 5$, random slopes, seed 2. Ambiguity peak: no SATA −17.0, SATA whole band −43.2, per sub-band −43.3 dB (improvement 26.2 dB).

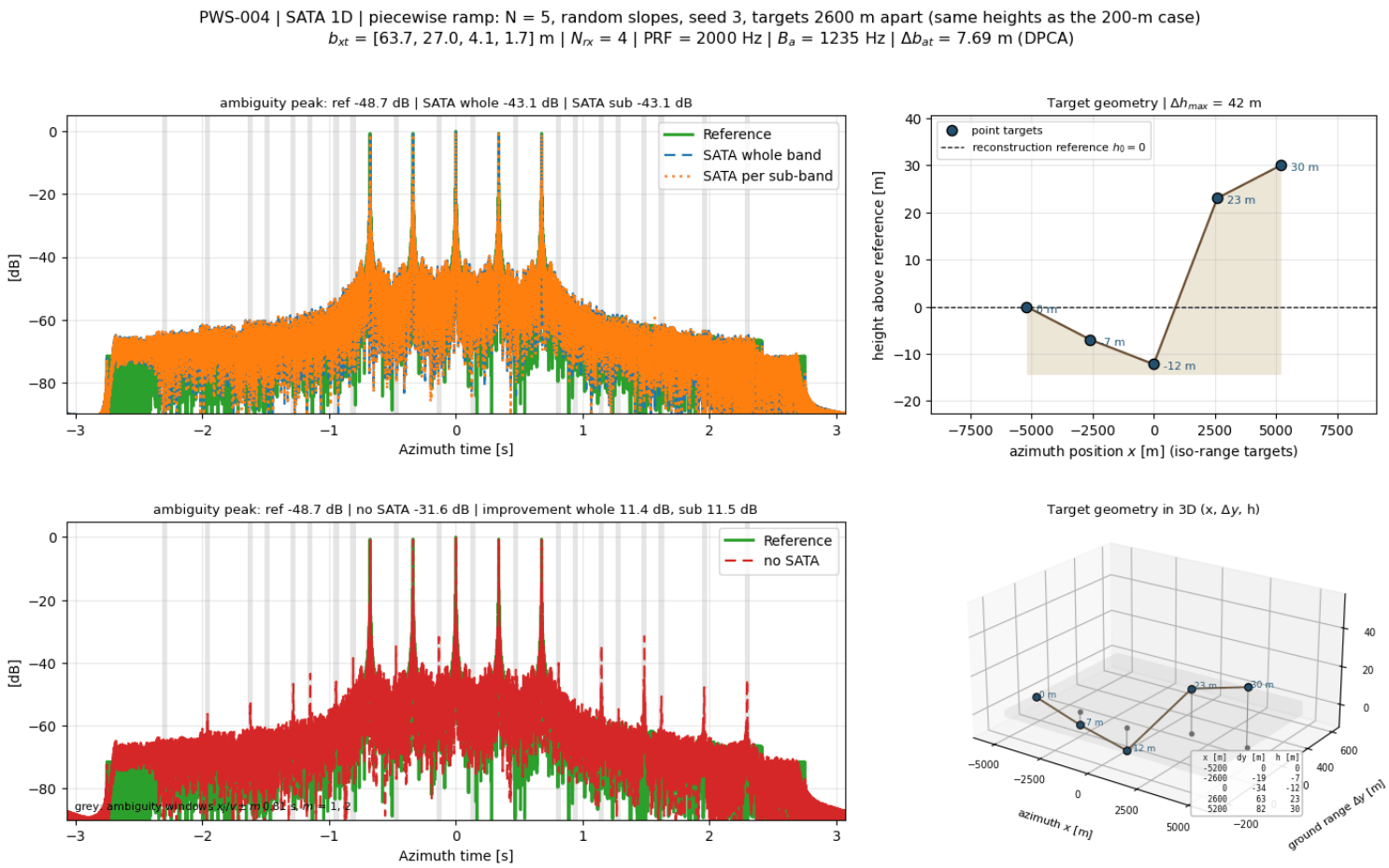


Figure 53: Piecewise ramp, 2.6 km apart, same heights: PWS-004, $N = 5$, random slopes, seed 3. Ambiguity peak: no SATA −31.6, SATA whole band −43.1, per sub-band −43.1 dB (improvement 11.4 dB).

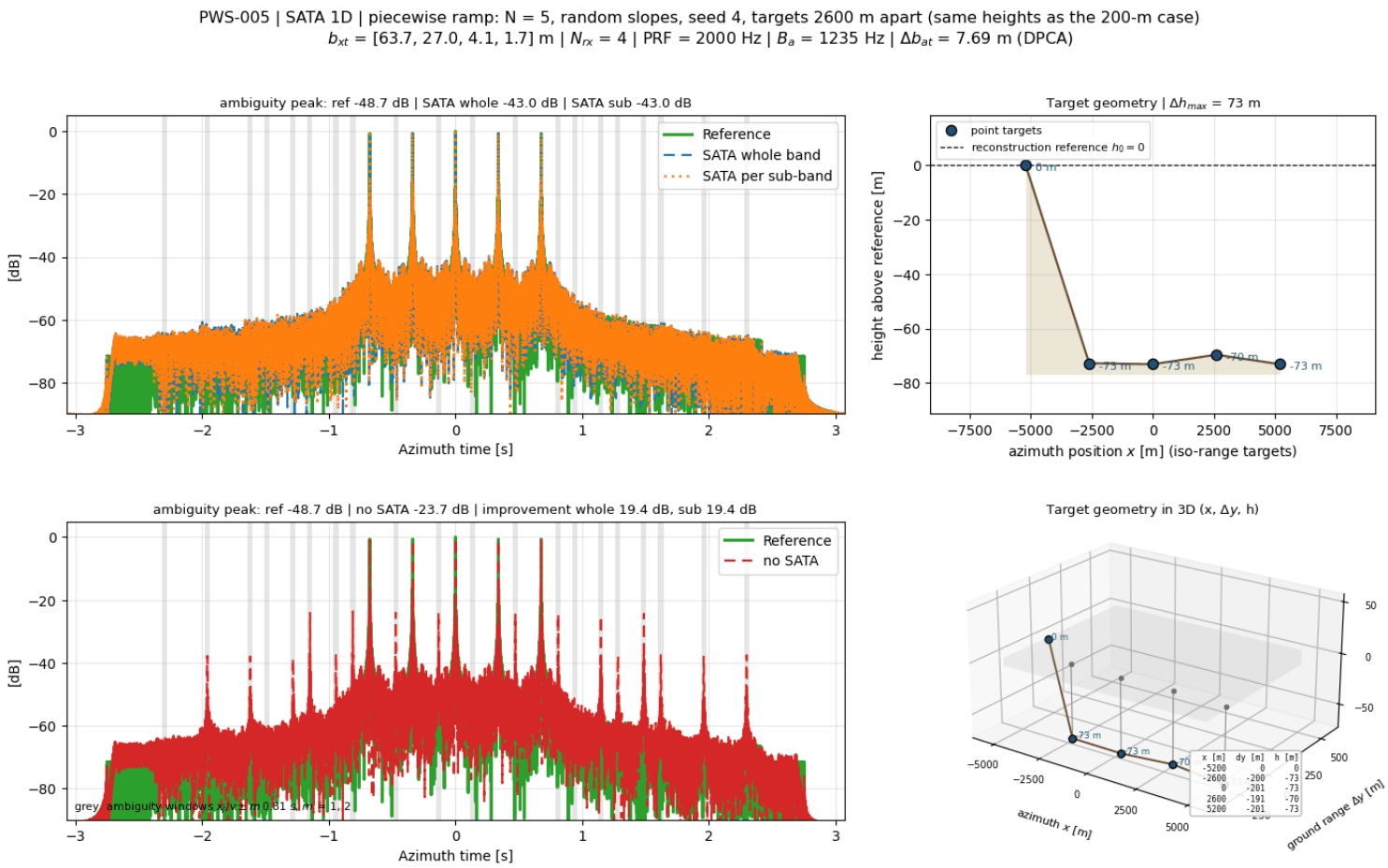


Figure 54: Piecewise ramp, 2.6 km apart, same heights: PWS-005, $N = 5$, random slopes, seed 4. Ambiguity peak: no SATA −23.7, SATA whole band −43.0, per sub-band −43.0 dB (improvement 19.4 dB).